Material Characterization Report

NanoVNA UTN Toolkit

2026-10-24 13:06:06

Measurement Details:

- **Sample:** Muestra1-22c
- **Technique:** Open-ended coaxial probe (liquids)
- **Temperature:** 23.0 °C
- **Reference Liquids:** Distilled Water / Isopropyl Alcohol (IPA)
- **Probe:** 3 mm probe
- **Frequency Range:** 0.005–6.000 GHz, 1200 pts
- **Data sources:**
  - **Open:** kit:cal_sonda3mm_UBA_1200p
  - **Short:** kit:cal_sonda3mm_UBA_1200p (1 spike(s) replaced in the computation: 4885 MHz)
  - **Ref 1:** kit:cal_sonda3mm_UBA_1200p
  - **Ref 2:** kit:cal_sonda3mm_UBA_1200p
  - **DUT:** imported

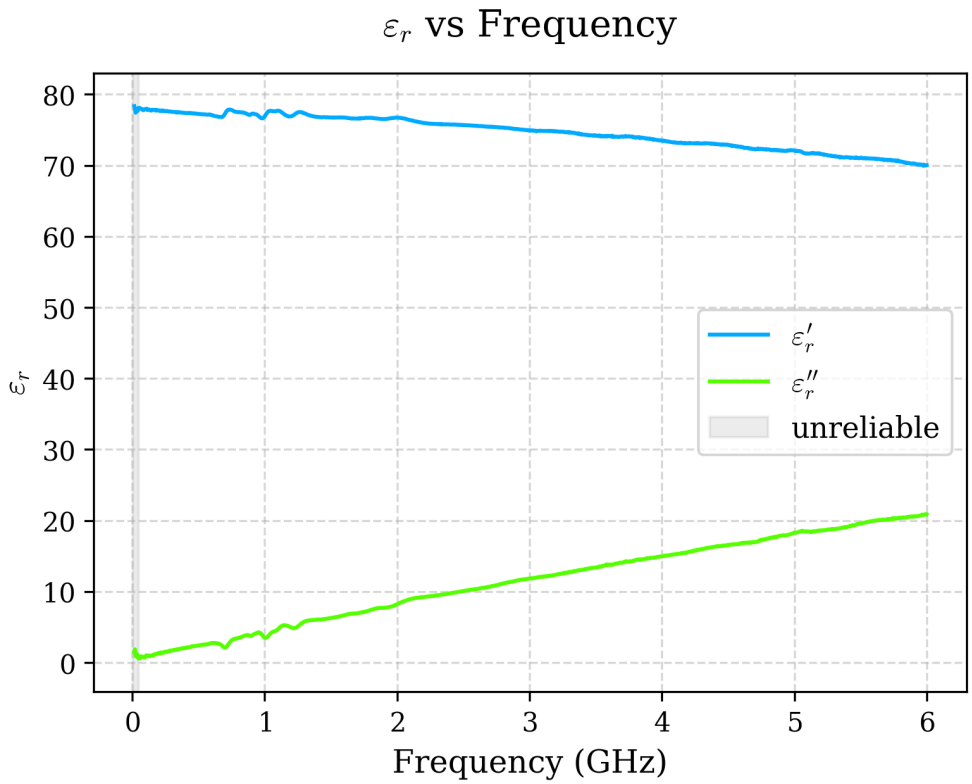
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1 Measurement Results

1.1 Complex Permittivity $\varepsilon_r(f)$



Points without a solution: 2 of 1200 (left blank in the plot and shown as — in the table).

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.005	—	—	—
0.010	—	—	—
0.015	78.3372	1.4891	0.0190
0.020	77.8851	1.8642	0.0239
0.025	77.4200	0.9000	0.0116
0.030	77.9465	1.3348	0.0171
0.035	77.8049	0.8174	0.0105
0.040	77.6604	1.0738	0.0138
0.045	78.0891	0.6129	0.0078
0.050	78.0481	0.8129	0.0104
0.055	78.0988	0.5584	0.0072
0.060	77.9839	0.8400	0.0108
0.065	77.9922	0.8793	0.0113
0.070	77.9327	0.8827	0.0113
0.075	77.8561	0.7781	0.0100
0.080	77.9078	0.8168	0.0105
0.085	77.7717	0.7610	0.0098

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.090	77.8998	0.7897	0.0101
0.095	77.9273	0.7204	0.0092
0.100	77.9239	0.8182	0.0105
0.105	77.8712	1.0230	0.0131
0.110	77.9896	0.9936	0.0127
0.115	77.8100	1.0174	0.0131
0.120	77.8694	0.9380	0.0120
0.125	77.8816	0.9821	0.0126
0.130	77.7883	0.8897	0.0114
0.135	77.7696	0.9031	0.0116
0.140	77.8210	0.9746	0.0125
0.145	77.7234	0.9973	0.0128
0.150	77.8330	1.0208	0.0131
0.155	77.8226	0.9413	0.0121
0.160	77.8174	1.1344	0.0146
0.165	77.8184	1.1209	0.0144
0.170	77.7757	1.1240	0.0145
0.175	77.8236	1.2371	0.0159
0.180	77.7424	1.1526	0.0148
0.185	77.8057	1.2308	0.0158
0.190	77.7516	1.3047	0.0168
0.195	77.7438	1.2969	0.0167
0.200	77.7148	1.3418	0.0173
0.205	77.7456	1.2338	0.0159
0.210	77.6593	1.4020	0.0181
0.215	77.6706	1.3338	0.0172
0.220	77.6584	1.4229	0.0183
0.225	77.7446	1.3794	0.0177
0.230	77.6933	1.3732	0.0177
0.235	77.7190	1.4006	0.0180
0.240	77.6758	1.4516	0.0187
0.245	77.6309	1.5334	0.0198
0.250	77.6454	1.5552	0.0200
0.255	77.6183	1.4865	0.0192
0.260	77.6698	1.5212	0.0196
0.265	77.6401	1.5571	0.0201
0.270	77.6005	1.5779	0.0203
0.275	77.6087	1.5803	0.0204
0.280	77.5850	1.6689	0.0215
0.285	77.5755	1.6129	0.0208
0.290	77.5891	1.7006	0.0219
0.295	77.5654	1.6849	0.0217
0.300	77.5423	1.7137	0.0221
0.305	77.5413	1.7472	0.0225
0.310	77.5443	1.7682	0.0228
0.315	77.5344	1.7453	0.0225

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.320	77.5140	1.7782	0.0229
0.325	77.4934	1.8181	0.0235
0.330	77.5240	1.8489	0.0238
0.335	77.4473	1.8106	0.0234
0.340	77.4461	1.8549	0.0240
0.345	77.5158	1.8376	0.0237
0.350	77.4888	1.9054	0.0246
0.355	77.4777	1.9346	0.0250
0.360	77.4755	1.9254	0.0249
0.365	77.4513	1.9475	0.0251
0.370	77.4456	1.9651	0.0254
0.375	77.4407	2.0248	0.0261
0.380	77.4312	2.0233	0.0261
0.385	77.4105	2.0019	0.0259
0.390	77.4173	2.0375	0.0263
0.395	77.3943	2.0536	0.0265
0.400	77.3744	2.0830	0.0269
0.405	77.3687	2.0783	0.0269
0.410	77.3732	2.1352	0.0276
0.415	77.3764	2.1117	0.0273
0.420	77.3639	2.1203	0.0274
0.425	77.3864	2.1782	0.0281
0.430	77.3982	2.1754	0.0281
0.435	77.3411	2.1806	0.0282
0.440	77.3660	2.2056	0.0285
0.445	77.3720	2.2363	0.0289
0.450	77.3274	2.3016	0.0298
0.455	77.3232	2.2790	0.0295
0.460	77.3310	2.3037	0.0298
0.465	77.2970	2.3247	0.0301
0.470	77.3152	2.3218	0.0300
0.475	77.3107	2.3386	0.0302
0.480	77.2845	2.3604	0.0305
0.485	77.2859	2.3632	0.0306
0.490	77.2646	2.3798	0.0308
0.495	77.2660	2.4184	0.0313
0.500	77.2366	2.4161	0.0313
0.505	77.2509	2.4149	0.0313
0.510	77.2363	2.4381	0.0316
0.515	77.2245	2.4566	0.0318
0.520	77.2355	2.4782	0.0321
0.525	77.1979	2.4736	0.0320
0.530	77.2046	2.5203	0.0326
0.535	77.2044	2.4915	0.0323
0.540	77.2006	2.5146	0.0326
0.545	77.1785	2.5569	0.0331

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.550	77.1845	2.5421	0.0329
0.555	77.1619	2.5693	0.0333
0.560	77.1562	2.5567	0.0331
0.565	77.1582	2.5848	0.0335
0.570	77.1455	2.6053	0.0338
0.575	77.1607	2.6109	0.0338
0.580	77.1678	2.6519	0.0344
0.585	77.1750	2.6523	0.0344
0.590	77.1267	2.7168	0.0352
0.595	77.1260	2.7308	0.0354
0.600	77.0938	2.7495	0.0357
0.605	77.0635	2.7588	0.0358
0.610	77.0254	2.7590	0.0358
0.615	76.9844	2.7566	0.0358
0.620	76.9701	2.7432	0.0356
0.625	76.9483	2.7563	0.0358
0.630	76.9243	2.7328	0.0355
0.635	76.9165	2.7277	0.0355
0.640	76.9048	2.7035	0.0352
0.645	76.8683	2.7156	0.0353
0.650	76.8683	2.6930	0.0350
0.655	76.8345	2.6477	0.0345
0.660	76.8274	2.6064	0.0339
0.665	76.8196	2.5719	0.0335
0.670	76.7939	2.5024	0.0326
0.675	76.8039	2.4286	0.0316
0.680	76.8424	2.3476	0.0306
0.685	76.8951	2.2530	0.0293
0.690	76.9977	2.1883	0.0284
0.695	77.1325	2.1132	0.0274
0.700	77.2716	2.1103	0.0273
0.705	77.4344	2.1413	0.0277
0.710	77.5938	2.1960	0.0283
0.715	77.6920	2.3228	0.0299
0.720	77.7912	2.4359	0.0313
0.725	77.8293	2.5725	0.0331
0.730	77.8567	2.7023	0.0347
0.735	77.8761	2.8036	0.0360
0.740	77.8472	2.9341	0.0377
0.745	77.8285	3.0513	0.0392
0.750	77.7882	3.1389	0.0404
0.755	77.7586	3.2008	0.0412
0.760	77.7037	3.2218	0.0415
0.765	77.6524	3.3020	0.0425
0.770	77.6140	3.3418	0.0431
0.775	77.5679	3.3260	0.0429

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.780	77.5717	3.3654	0.0434
0.785	77.5354	3.3888	0.0437
0.790	77.5181	3.3924	0.0438
0.795	77.5112	3.4314	0.0443
0.800	77.5245	3.4640	0.0447
0.805	77.5014	3.4673	0.0447
0.810	77.4944	3.5110	0.0453
0.815	77.5034	3.5522	0.0458
0.820	77.4735	3.5793	0.0462
0.825	77.4647	3.6035	0.0465
0.830	77.4660	3.6393	0.0470
0.835	77.4345	3.6841	0.0476
0.840	77.4322	3.7196	0.0480
0.845	77.4158	3.7672	0.0487
0.850	77.3730	3.7972	0.0491
0.855	77.3506	3.8281	0.0495
0.860	77.2829	3.8363	0.0496
0.865	77.2657	3.8776	0.0502
0.870	77.2139	3.8865	0.0503
0.875	77.1638	3.8716	0.0502
0.880	77.1235	3.8474	0.0499
0.885	77.1205	3.8028	0.0493
0.890	77.1228	3.7818	0.0490
0.895	77.1863	3.7399	0.0485
0.900	77.2173	3.7537	0.0486
0.905	77.2892	3.8137	0.0493
0.910	77.2978	3.8861	0.0503
0.915	77.2827	3.9610	0.0513
0.920	77.2516	4.0104	0.0519
0.925	77.2305	4.0539	0.0525
0.930	77.2083	4.0826	0.0529
0.935	77.1821	4.1052	0.0532
0.940	77.1568	4.1676	0.0540
0.945	77.0940	4.2199	0.0547
0.950	77.0174	4.2466	0.0551
0.955	76.9252	4.2589	0.0554
0.960	76.8408	4.2115	0.0548
0.965	76.7790	4.1941	0.0546
0.970	76.6980	4.1174	0.0537
0.975	76.6402	4.0118	0.0523
0.980	76.6332	3.9074	0.0510
0.985	76.6432	3.7621	0.0491
0.990	76.7148	3.6725	0.0479
0.995	76.7815	3.5500	0.0462
1.000	76.9207	3.4779	0.0452
1.005	77.0762	3.4732	0.0451

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.010	77.2142	3.4885	0.0452
1.015	77.3399	3.5302	0.0456
1.020	77.4515	3.6090	0.0466
1.025	77.5637	3.6828	0.0475
1.030	77.6236	3.8116	0.0491
1.035	77.6633	3.9011	0.0502
1.040	77.6886	4.0241	0.0518
1.045	77.6814	4.1091	0.0529
1.050	77.6728	4.1977	0.0540
1.055	77.6426	4.2559	0.0548
1.060	77.6473	4.3304	0.0558
1.065	77.6206	4.3599	0.0562
1.070	77.6175	4.3804	0.0564
1.075	77.6091	4.4318	0.0571
1.080	77.6184	4.4625	0.0575
1.085	77.6369	4.4884	0.0578
1.090	77.6776	4.5350	0.0584
1.095	77.6935	4.6207	0.0595
1.100	77.7053	4.6990	0.0605
1.105	77.6937	4.8015	0.0618
1.110	77.6748	4.8788	0.0628
1.115	77.6442	4.9658	0.0640
1.120	77.5833	5.0643	0.0653
1.125	77.5136	5.1228	0.0661
1.130	77.4321	5.1655	0.0667
1.135	77.3921	5.1987	0.0672
1.140	77.3169	5.2380	0.0677
1.145	77.2375	5.2751	0.0683
1.150	77.1485	5.2484	0.0680
1.155	77.1234	5.2477	0.0680
1.160	77.0548	5.2319	0.0679
1.165	77.0060	5.2279	0.0679
1.170	76.9539	5.1737	0.0672
1.175	76.9381	5.1553	0.0670
1.180	76.8937	5.1168	0.0665
1.185	76.8923	5.0756	0.0660
1.190	76.8935	5.0258	0.0654
1.195	76.8980	4.9973	0.0650
1.200	76.9301	4.9537	0.0644
1.205	76.9473	4.9234	0.0640
1.210	76.9722	4.8790	0.0634
1.215	77.0395	4.8617	0.0631
1.220	77.1076	4.8490	0.0629
1.225	77.1607	4.8588	0.0630
1.230	77.2340	4.8875	0.0633
1.235	77.2964	4.8980	0.0634

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.240	77.3761	4.9421	0.0639
1.245	77.4280	5.0089	0.0647
1.250	77.4525	5.1011	0.0659
1.255	77.4948	5.1723	0.0667
1.260	77.5171	5.2332	0.0675
1.265	77.5087	5.3111	0.0685
1.270	77.5127	5.3897	0.0695
1.275	77.4891	5.4816	0.0707
1.280	77.4728	5.5583	0.0717
1.285	77.4539	5.6070	0.0724
1.290	77.3880	5.6656	0.0732
1.295	77.3728	5.7291	0.0740
1.300	77.3393	5.7685	0.0746
1.305	77.2868	5.8178	0.0753
1.310	77.2782	5.8446	0.0756
1.315	77.2152	5.8673	0.0760
1.320	77.1691	5.8831	0.0762
1.325	77.1355	5.9151	0.0767
1.330	77.1093	5.9404	0.0770
1.335	77.0724	5.9376	0.0770
1.340	77.0399	5.9453	0.0772
1.345	77.0155	5.9545	0.0773
1.350	76.9879	5.9922	0.0778
1.355	76.9710	5.9780	0.0777
1.360	76.9486	5.9848	0.0778
1.365	76.9087	5.9867	0.0778
1.370	76.8952	5.9949	0.0780
1.375	76.8848	6.0010	0.0781
1.380	76.8711	6.0165	0.0783
1.385	76.8591	6.0336	0.0785
1.390	76.8357	6.0473	0.0787
1.395	76.8354	6.0257	0.0784
1.400	76.8233	6.0588	0.0789
1.405	76.8115	6.0457	0.0787
1.410	76.7969	6.0657	0.0790
1.415	76.7919	6.0568	0.0789
1.420	76.7865	6.0539	0.0788
1.425	76.7806	6.0538	0.0788
1.430	76.7820	6.0880	0.0793
1.435	76.7672	6.0843	0.0793
1.440	76.7917	6.0895	0.0793
1.445	76.7994	6.1103	0.0796
1.450	76.7908	6.1247	0.0798
1.455	76.7723	6.1443	0.0800
1.460	76.7642	6.1578	0.0802
1.465	76.7795	6.1789	0.0805

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.470	76.7770	6.1820	0.0805
1.475	76.7615	6.1903	0.0806
1.480	76.7648	6.2309	0.0812
1.485	76.7578	6.2298	0.0812
1.490	76.7538	6.2508	0.0814
1.495	76.7457	6.2724	0.0817
1.500	76.7529	6.2715	0.0817
1.505	76.7519	6.2859	0.0819
1.510	76.7357	6.3098	0.0822
1.515	76.7405	6.3165	0.0823
1.520	76.7558	6.3341	0.0825
1.525	76.7230	6.3660	0.0830
1.530	76.7462	6.3654	0.0829
1.535	76.7447	6.3937	0.0833
1.540	76.7425	6.4095	0.0835
1.545	76.7479	6.4141	0.0836
1.550	76.7563	6.4480	0.0840
1.555	76.7397	6.4560	0.0841
1.560	76.7625	6.4853	0.0845
1.565	76.7578	6.5116	0.0848
1.570	76.7468	6.5276	0.0851
1.575	76.7405	6.5278	0.0851
1.580	76.7673	6.5731	0.0856
1.585	76.7402	6.6129	0.0862
1.590	76.7404	6.6244	0.0863
1.595	76.7329	6.6398	0.0865
1.600	76.7345	6.6684	0.0869
1.605	76.7446	6.6834	0.0871
1.610	76.7157	6.7126	0.0875
1.615	76.7130	6.7387	0.0878
1.620	76.6954	6.7799	0.0884
1.625	76.6836	6.7960	0.0886
1.630	76.6754	6.8122	0.0888
1.635	76.6716	6.8356	0.0892
1.640	76.6578	6.8283	0.0891
1.645	76.6456	6.8422	0.0893
1.650	76.6035	6.8622	0.0896
1.655	76.6028	6.8889	0.0899
1.660	76.6049	6.8808	0.0898
1.665	76.6245	6.8848	0.0899
1.670	76.6064	6.9041	0.0901
1.675	76.6064	6.9250	0.0904
1.680	76.6031	6.9338	0.0905
1.685	76.6018	6.9220	0.0904
1.690	76.5970	6.9374	0.0906
1.695	76.6137	6.9493	0.0907

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.700	76.5963	6.9553	0.0908
1.705	76.5946	6.9878	0.0912
1.710	76.6282	6.9754	0.0910
1.715	76.5997	6.9891	0.0912
1.720	76.6325	7.0315	0.0918
1.725	76.6082	7.0475	0.0920
1.730	76.6568	7.0708	0.0922
1.735	76.6429	7.0710	0.0923
1.740	76.6464	7.0907	0.0925
1.745	76.6505	7.1362	0.0931
1.750	76.6686	7.1163	0.0928
1.755	76.6772	7.1468	0.0932
1.760	76.6888	7.1590	0.0934
1.765	76.7042	7.1764	0.0936
1.770	76.6995	7.2305	0.0943
1.775	76.6999	7.2493	0.0945
1.780	76.6932	7.2841	0.0950
1.785	76.7013	7.3016	0.0952
1.790	76.6906	7.3209	0.0955
1.795	76.6935	7.3777	0.0962
1.800	76.7002	7.4033	0.0965
1.805	76.7035	7.4329	0.0969
1.810	76.6907	7.4618	0.0973
1.815	76.6968	7.4697	0.0974
1.820	76.6601	7.5125	0.0980
1.825	76.6549	7.5322	0.0983
1.830	76.6479	7.5764	0.0988
1.835	76.6262	7.5843	0.0990
1.840	76.6209	7.6244	0.0995
1.845	76.6207	7.6545	0.0999
1.850	76.6059	7.6352	0.0997
1.855	76.5874	7.6561	0.1000
1.860	76.5813	7.6571	0.1000
1.865	76.5626	7.7028	0.1006
1.870	76.5618	7.7110	0.1007
1.875	76.5424	7.7066	0.1007
1.880	76.5325	7.6946	0.1005
1.885	76.5187	7.6996	0.1006
1.890	76.5384	7.7101	0.1007
1.895	76.5133	7.6987	0.1006
1.900	76.5195	7.7045	0.1007
1.905	76.5368	7.7233	0.1009
1.910	76.5205	7.7407	0.1012
1.915	76.5153	7.7561	0.1014
1.920	76.5494	7.7517	0.1013
1.925	76.5670	7.7450	0.1012

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.930	76.5738	7.7700	0.1015
1.935	76.5851	7.7934	0.1018
1.940	76.6063	7.7934	0.1017
1.945	76.6312	7.8388	0.1023
1.950	76.6368	7.8482	0.1024
1.955	76.6452	7.8834	0.1029
1.960	76.6583	7.9001	0.1031
1.965	76.6779	7.9355	0.1035
1.970	76.6760	7.9650	0.1039
1.975	76.6917	8.0275	0.1047
1.980	76.6976	8.0449	0.1049
1.985	76.7256	8.0630	0.1051
1.990	76.7271	8.1387	0.1061
1.995	76.7192	8.1702	0.1065
2.000	76.7350	8.2127	0.1070
2.005	76.7187	8.2587	0.1076
2.010	76.7217	8.2959	0.1081
2.015	76.7153	8.3337	0.1086
2.020	76.7093	8.3977	0.1095
2.025	76.6854	8.4428	0.1101
2.030	76.6751	8.4659	0.1104
2.035	76.6676	8.5188	0.1111
2.040	76.6361	8.5510	0.1116
2.045	76.6318	8.5950	0.1122
2.050	76.6210	8.6200	0.1125
2.055	76.5947	8.6680	0.1132
2.060	76.5622	8.6923	0.1135
2.065	76.5723	8.7199	0.1139
2.070	76.5256	8.7545	0.1144
2.075	76.4931	8.7930	0.1150
2.080	76.4661	8.8197	0.1153
2.085	76.4437	8.8652	0.1160
2.090	76.4242	8.8855	0.1163
2.095	76.4084	8.9203	0.1167
2.100	76.3706	8.9242	0.1169
2.105	76.3433	8.9728	0.1175
2.110	76.3401	8.9820	0.1177
2.115	76.3130	9.0109	0.1181
2.120	76.2948	9.0126	0.1181
2.125	76.2760	9.0267	0.1183
2.130	76.2309	9.0544	0.1188
2.135	76.2185	9.0550	0.1188
2.140	76.1952	9.1020	0.1195
2.145	76.1726	9.1250	0.1198
2.150	76.1821	9.1151	0.1196
2.155	76.1437	9.1292	0.1199

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.160	76.1087	9.1411	0.1201
2.165	76.1071	9.1531	0.1203
2.170	76.0819	9.1595	0.1204
2.175	76.0747	9.1854	0.1207
2.180	76.0459	9.1937	0.1209
2.185	76.0378	9.1954	0.1209
2.190	76.0079	9.2432	0.1216
2.195	75.9845	9.2214	0.1214
2.200	75.9822	9.2343	0.1215
2.205	75.9637	9.2418	0.1217
2.210	75.9670	9.2581	0.1219
2.215	75.9402	9.2820	0.1222
2.220	75.9474	9.2790	0.1222
2.225	75.9344	9.2834	0.1223
2.230	75.9070	9.2979	0.1225
2.235	75.8980	9.3042	0.1226
2.240	75.8928	9.3319	0.1230
2.245	75.8776	9.3357	0.1230
2.250	75.8806	9.3543	0.1233
2.255	75.8622	9.3612	0.1234
2.260	75.8603	9.3721	0.1235
2.265	75.8547	9.3653	0.1235
2.270	75.8618	9.3963	0.1239
2.275	75.8439	9.3757	0.1236
2.280	75.8408	9.4048	0.1240
2.285	75.8352	9.4282	0.1243
2.290	75.8279	9.4403	0.1245
2.295	75.8340	9.4628	0.1248
2.300	75.8193	9.4570	0.1247
2.305	75.8249	9.4985	0.1253
2.310	75.8059	9.4854	0.1251
2.315	75.8336	9.4853	0.1251
2.320	75.8005	9.5102	0.1255
2.325	75.8078	9.5101	0.1255
2.330	75.8303	9.5477	0.1259
2.335	75.7742	9.5681	0.1263
2.340	75.8093	9.5605	0.1261
2.345	75.7827	9.5793	0.1264
2.350	75.7810	9.6051	0.1267
2.355	75.7858	9.6130	0.1268
2.360	75.7657	9.6121	0.1269
2.365	75.7761	9.6585	0.1275
2.370	75.7917	9.6418	0.1272
2.375	75.7703	9.6792	0.1277
2.380	75.7903	9.6723	0.1276
2.385	75.7822	9.7031	0.1280

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.390	75.7734	9.7114	0.1282
2.395	75.7646	9.7409	0.1286
2.400	75.7654	9.7591	0.1288
2.405	75.7467	9.7585	0.1288
2.410	75.7665	9.7839	0.1291
2.415	75.7520	9.8088	0.1295
2.420	75.7530	9.8120	0.1295
2.425	75.7534	9.8327	0.1298
2.430	75.7462	9.8218	0.1297
2.435	75.7472	9.8561	0.1301
2.440	75.7551	9.8768	0.1304
2.445	75.7263	9.8978	0.1307
2.450	75.7368	9.9001	0.1307
2.455	75.7179	9.9186	0.1310
2.460	75.7294	9.9460	0.1313
2.465	75.7157	9.9682	0.1317
2.470	75.7029	9.9787	0.1318
2.475	75.7160	9.9902	0.1319
2.480	75.7009	10.0108	0.1322
2.485	75.6893	10.0339	0.1326
2.490	75.6919	10.0293	0.1325
2.495	75.7042	10.0602	0.1329
2.500	75.6800	10.0773	0.1332
2.505	75.6738	10.0955	0.1334
2.510	75.6684	10.1125	0.1336
2.515	75.6695	10.1092	0.1336
2.520	75.6575	10.1434	0.1341
2.525	75.6596	10.1641	0.1343
2.530	75.6481	10.1569	0.1343
2.535	75.6422	10.1942	0.1348
2.540	75.6367	10.2047	0.1349
2.545	75.6387	10.2062	0.1349
2.550	75.6396	10.2210	0.1351
2.555	75.6340	10.2545	0.1356
2.560	75.6160	10.2552	0.1356
2.565	75.6097	10.2827	0.1360
2.570	75.5944	10.2708	0.1359
2.575	75.6027	10.2993	0.1362
2.580	75.6024	10.3013	0.1363
2.585	75.5857	10.3294	0.1367
2.590	75.5718	10.3382	0.1368
2.595	75.5866	10.3694	0.1372
2.600	75.5799	10.3704	0.1372
2.605	75.5463	10.3682	0.1372
2.610	75.5690	10.3911	0.1375
2.615	75.5442	10.4149	0.1379

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.620	75.5445	10.4288	0.1380
2.625	75.5279	10.4403	0.1382
2.630	75.5401	10.4529	0.1384
2.635	75.5443	10.4767	0.1387
2.640	75.5149	10.4965	0.1390
2.645	75.5193	10.5130	0.1392
2.650	75.5061	10.5367	0.1395
2.655	75.5132	10.5361	0.1395
2.660	75.4899	10.5506	0.1398
2.665	75.4855	10.5894	0.1403
2.670	75.4692	10.5921	0.1404
2.675	75.4736	10.6016	0.1405
2.680	75.4611	10.6280	0.1408
2.685	75.4674	10.6443	0.1410
2.690	75.4326	10.6461	0.1411
2.695	75.4441	10.6670	0.1414
2.700	75.4517	10.6998	0.1418
2.705	75.4376	10.7052	0.1419
2.710	75.4686	10.7485	0.1424
2.715	75.4404	10.7477	0.1425
2.720	75.4263	10.7823	0.1430
2.725	75.4102	10.7912	0.1431
2.730	75.4245	10.8023	0.1432
2.735	75.4129	10.8455	0.1438
2.740	75.4166	10.8399	0.1437
2.745	75.3886	10.8771	0.1443
2.750	75.3795	10.8964	0.1446
2.755	75.3903	10.9234	0.1449
2.760	75.3896	10.9292	0.1450
2.765	75.3510	10.9709	0.1456
2.770	75.3411	11.0008	0.1460
2.775	75.3530	11.0204	0.1463
2.780	75.3378	11.0313	0.1464
2.785	75.3498	11.0427	0.1466
2.790	75.3304	11.0734	0.1470
2.795	75.3260	11.1209	0.1476
2.800	75.3315	11.1416	0.1479
2.805	75.3043	11.1589	0.1482
2.810	75.2777	11.1741	0.1484
2.815	75.2998	11.2209	0.1490
2.820	75.2883	11.1962	0.1487
2.825	75.2749	11.2178	0.1490
2.830	75.2780	11.2741	0.1498
2.835	75.2815	11.2721	0.1497
2.840	75.2647	11.3113	0.1503
2.845	75.2346	11.3192	0.1505

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.850	75.2391	11.3398	0.1507
2.855	75.2195	11.3609	0.1510
2.860	75.2258	11.3750	0.1512
2.865	75.1913	11.3846	0.1514
2.870	75.1634	11.4025	0.1517
2.875	75.1742	11.4269	0.1520
2.880	75.1670	11.4344	0.1521
2.885	75.1381	11.4773	0.1527
2.890	75.1334	11.4896	0.1529
2.895	75.1165	11.5263	0.1534
2.900	75.1300	11.5202	0.1533
2.905	75.0964	11.5485	0.1538
2.910	75.1060	11.5792	0.1542
2.915	75.1013	11.6060	0.1545
2.920	75.0873	11.6216	0.1548
2.925	75.0722	11.6243	0.1548
2.930	75.0433	11.6543	0.1553
2.935	75.0491	11.6620	0.1554
2.940	75.0566	11.6753	0.1556
2.945	75.0287	11.6984	0.1559
2.950	75.0195	11.6988	0.1559
2.955	75.0255	11.6904	0.1558
2.960	75.0128	11.7390	0.1565
2.965	74.9884	11.7222	0.1563
2.970	74.9821	11.7304	0.1564
2.975	74.9793	11.7546	0.1568
2.980	74.9844	11.7678	0.1569
2.985	74.9371	11.8092	0.1576
2.990	74.9517	11.8018	0.1575
2.995	74.9209	11.7919	0.1574
3.000	74.9446	11.8329	0.1579
3.005	74.8801	11.8647	0.1584
3.010	74.9192	11.8530	0.1582
3.015	74.9222	11.8525	0.1582
3.020	74.8998	11.9000	0.1589
3.025	74.9308	11.8498	0.1581
3.030	74.9001	11.8792	0.1586
3.035	74.8971	11.9010	0.1589
3.040	74.9027	11.9572	0.1596
3.045	74.8716	11.9219	0.1592
3.050	74.7957	11.9305	0.1595
3.055	74.8146	11.9895	0.1603
3.060	74.8651	12.0030	0.1603
3.065	74.8640	12.0076	0.1604
3.070	74.8759	12.0241	0.1606
3.075	74.8821	12.0607	0.1611

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.080	74.8508	12.0439	0.1609
3.085	74.8980	12.0785	0.1613
3.090	74.8664	12.0869	0.1614
3.095	74.8608	12.0920	0.1615
3.100	74.8637	12.1054	0.1617
3.105	74.8453	12.0837	0.1614
3.110	74.8492	12.1363	0.1621
3.115	74.8541	12.1517	0.1623
3.120	74.8464	12.1255	0.1620
3.125	74.8411	12.1358	0.1622
3.130	74.8422	12.1280	0.1620
3.135	74.8581	12.1812	0.1627
3.140	74.8605	12.2063	0.1631
3.145	74.8254	12.1645	0.1626
3.150	74.8667	12.2099	0.1631
3.155	74.8349	12.2512	0.1637
3.160	74.8570	12.2477	0.1636
3.165	74.8365	12.2482	0.1637
3.170	74.8518	12.2496	0.1637
3.175	74.8273	12.2973	0.1643
3.180	74.8442	12.2957	0.1643
3.185	74.7839	12.3190	0.1647
3.190	74.8145	12.3471	0.1650
3.195	74.8168	12.3613	0.1652
3.200	74.7715	12.3725	0.1655
3.205	74.8033	12.4029	0.1658
3.210	74.7918	12.4229	0.1661
3.215	74.7616	12.4350	0.1663
3.220	74.7666	12.4342	0.1663
3.225	74.7592	12.4725	0.1668
3.230	74.7340	12.4614	0.1667
3.235	74.7487	12.4957	0.1672
3.240	74.7515	12.5288	0.1676
3.245	74.7602	12.5523	0.1679
3.250	74.7361	12.5550	0.1680
3.255	74.7412	12.6011	0.1686
3.260	74.7388	12.5873	0.1684
3.265	74.6872	12.6607	0.1695
3.270	74.6579	12.6261	0.1691
3.275	74.6745	12.6573	0.1695
3.280	74.6765	12.6423	0.1693
3.285	74.6758	12.6996	0.1701
3.290	74.6835	12.6307	0.1691
3.295	74.6259	12.7131	0.1704
3.300	74.6333	12.7233	0.1705
3.305	74.6916	12.7911	0.1713

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.310	74.6200	12.7371	0.1707
3.315	74.5848	12.7520	0.1710
3.320	74.6181	12.7900	0.1714
3.325	74.6171	12.7575	0.1710
3.330	74.5988	12.8353	0.1721
3.335	74.5688	12.8889	0.1728
3.340	74.5827	12.8793	0.1727
3.345	74.5279	12.8562	0.1725
3.350	74.5166	12.8971	0.1731
3.355	74.5020	12.9240	0.1735
3.360	74.4794	12.9477	0.1738
3.365	74.5350	12.9765	0.1741
3.370	74.4254	13.0216	0.1750
3.375	74.4851	12.9824	0.1743
3.380	74.4220	13.0078	0.1748
3.385	74.3496	13.0276	0.1752
3.390	74.4216	13.0555	0.1754
3.395	74.3865	13.0336	0.1752
3.400	74.3596	13.0569	0.1756
3.405	74.3397	13.0716	0.1758
3.410	74.3321	13.1083	0.1763
3.415	74.3338	13.1120	0.1764
3.420	74.3063	13.1764	0.1773
3.425	74.2965	13.1765	0.1774
3.430	74.3017	13.1616	0.1771
3.435	74.3190	13.2217	0.1779
3.440	74.2669	13.2323	0.1782
3.445	74.2461	13.2013	0.1778
3.450	74.2142	13.2438	0.1785
3.455	74.2677	13.2801	0.1788
3.460	74.2583	13.2860	0.1789
3.465	74.2335	13.3362	0.1797
3.470	74.2106	13.2920	0.1791
3.475	74.2283	13.3205	0.1795
3.480	74.2156	13.3421	0.1798
3.485	74.1681	13.3565	0.1801
3.490	74.2417	13.3640	0.1800
3.495	74.1790	13.3738	0.1803
3.500	74.2063	13.3851	0.1804
3.505	74.2209	13.3680	0.1801
3.510	74.1781	13.3959	0.1806
3.515	74.1490	13.4252	0.1811
3.520	74.1875	13.4805	0.1817
3.525	74.1152	13.4514	0.1815
3.530	74.2273	13.5368	0.1824
3.535	74.1551	13.4818	0.1818

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.540	74.0771	13.5378	0.1828
3.545	74.1438	13.5358	0.1826
3.550	74.0880	13.6293	0.1840
3.555	74.1122	13.5788	0.1832
3.560	74.0916	13.6397	0.1841
3.565	74.1275	13.6663	0.1844
3.570	74.1160	13.7030	0.1849
3.575	74.0981	13.7625	0.1857
3.580	74.1772	13.7051	0.1848
3.585	74.0756	13.8270	0.1867
3.590	74.1385	13.7637	0.1856
3.595	74.1396	13.7653	0.1857
3.600	74.1511	13.7819	0.1859
3.605	74.1153	13.7940	0.1861
3.610	74.1103	13.7657	0.1857
3.615	74.0667	13.7996	0.1863
3.620	74.0686	13.7624	0.1858
3.625	74.1979	13.7823	0.1858
3.630	74.0005	13.7897	0.1863
3.635	74.0341	13.8280	0.1868
3.640	74.0418	13.8382	0.1869
3.645	74.0648	13.9192	0.1879
3.650	74.0345	13.9383	0.1883
3.655	74.0496	13.9290	0.1881
3.660	74.0318	13.9632	0.1886
3.665	74.0421	13.9658	0.1886
3.670	74.0133	13.9954	0.1891
3.675	74.0136	14.0287	0.1895
3.680	74.0044	14.0159	0.1894
3.685	74.0504	14.0569	0.1898
3.690	74.0738	14.0821	0.1901
3.695	74.0711	14.0751	0.1900
3.700	74.0886	14.0664	0.1899
3.705	74.0284	14.0848	0.1903
3.710	74.0539	14.1343	0.1909
3.715	74.0592	14.1324	0.1908
3.720	74.0576	14.1197	0.1907
3.725	74.1519	14.2705	0.1925
3.730	74.0648	14.2004	0.1917
3.735	74.0531	14.2254	0.1921
3.740	74.0711	14.2240	0.1920
3.745	74.0909	14.2121	0.1918
3.750	74.1125	14.2247	0.1919
3.755	74.0886	14.2652	0.1925
3.760	74.0596	14.2179	0.1920
3.765	74.0658	14.2551	0.1925

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.770	74.0583	14.2684	0.1927
3.775	74.0159	14.2814	0.1930
3.780	74.0632	14.3000	0.1931
3.785	73.9681	14.3517	0.1940
3.790	73.9714	14.3259	0.1937
3.795	73.9255	14.3929	0.1947
3.800	73.8927	14.4208	0.1952
3.805	73.9786	14.4494	0.1953
3.810	73.9468	14.4855	0.1959
3.815	73.9403	14.5198	0.1964
3.820	73.9610	14.5076	0.1962
3.825	73.9934	14.5154	0.1962
3.830	73.9566	14.5487	0.1967
3.835	73.8262	14.4316	0.1955
3.840	73.9160	14.5428	0.1967
3.845	73.9183	14.5369	0.1967
3.850	73.9015	14.5380	0.1967
3.855	73.8986	14.5280	0.1966
3.860	73.8554	14.5747	0.1973
3.865	73.8575	14.5705	0.1973
3.870	73.8421	14.5946	0.1976
3.875	73.8145	14.6120	0.1980
3.880	73.8340	14.6589	0.1985
3.885	73.7508	14.7003	0.1993
3.890	73.7822	14.6320	0.1983
3.895	73.7689	14.6411	0.1985
3.900	73.7604	14.6713	0.1989
3.905	73.7315	14.7077	0.1995
3.910	73.7710	14.7314	0.1997
3.915	73.7061	14.7611	0.2003
3.920	73.7174	14.7503	0.2001
3.925	73.7092	14.7463	0.2001
3.930	73.6934	14.7648	0.2004
3.935	73.6001	14.7560	0.2005
3.940	73.6725	14.7877	0.2007
3.945	73.6212	14.7657	0.2006
3.950	73.6223	14.8040	0.2011
3.955	73.6040	14.8281	0.2015
3.960	73.5717	14.8645	0.2020
3.965	73.5694	14.8697	0.2021
3.970	73.5761	14.9040	0.2026
3.975	73.5641	14.8974	0.2025
3.980	73.5119	14.9084	0.2028
3.985	73.5509	14.9331	0.2030
3.990	73.5104	14.9390	0.2032
3.995	73.5197	14.9547	0.2034

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.000	73.5060	14.9628	0.2036
4.005	73.4652	14.9613	0.2037
4.010	73.4385	14.9964	0.2042
4.015	73.4994	15.0258	0.2044
4.020	73.4250	15.0340	0.2048
4.025	73.4406	15.0190	0.2045
4.030	73.4077	15.0847	0.2055
4.035	73.3827	15.0534	0.2051
4.040	73.4024	15.0567	0.2051
4.045	73.3650	15.0881	0.2057
4.050	73.3324	15.1007	0.2059
4.055	73.3410	15.1050	0.2060
4.060	73.3321	15.1319	0.2063
4.065	73.3024	15.1495	0.2067
4.070	73.2880	15.1691	0.2070
4.075	73.2657	15.1817	0.2072
4.080	73.2624	15.2153	0.2077
4.085	73.2415	15.2062	0.2076
4.090	73.2518	15.2277	0.2079
4.095	73.2644	15.2411	0.2080
4.100	73.2127	15.2533	0.2083
4.105	73.1999	15.2577	0.2084
4.110	73.2038	15.2705	0.2086
4.115	73.1751	15.3205	0.2094
4.120	73.1886	15.3035	0.2091
4.125	73.1494	15.2958	0.2091
4.130	73.1880	15.3291	0.2094
4.135	73.2031	15.3323	0.2094
4.140	73.1760	15.3540	0.2098
4.145	73.2136	15.4103	0.2105
4.150	73.1583	15.3738	0.2101
4.155	73.1237	15.4158	0.2108
4.160	73.1430	15.4390	0.2111
4.165	73.1399	15.4575	0.2113
4.170	73.1092	15.4591	0.2115
4.175	73.1656	15.4126	0.2107
4.180	73.1232	15.4872	0.2118
4.185	73.1107	15.4892	0.2119
4.190	73.1080	15.5244	0.2123
4.195	73.1056	15.5431	0.2126
4.200	73.1426	15.5642	0.2128
4.205	73.1582	15.5562	0.2126
4.210	73.1094	15.6287	0.2138
4.215	73.1416	15.6404	0.2138
4.220	73.1312	15.6310	0.2137
4.225	73.1125	15.6669	0.2143

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.230	73.1095	15.6722	0.2144
4.235	73.1100	15.7324	0.2152
4.240	73.1247	15.7061	0.2148
4.245	73.1048	15.7744	0.2158
4.250	73.1286	15.7682	0.2156
4.255	73.0928	15.8007	0.2162
4.260	73.0915	15.7898	0.2160
4.265	73.1448	15.8624	0.2169
4.270	73.1027	15.8404	0.2167
4.275	73.1393	15.8538	0.2168
4.280	73.0955	15.8843	0.2173
4.285	73.1039	15.9128	0.2177
4.290	73.0765	15.9490	0.2183
4.295	73.0857	15.9357	0.2180
4.300	73.0815	16.0021	0.2190
4.305	73.0638	15.9786	0.2187
4.310	73.0990	15.9928	0.2188
4.315	73.0948	16.0320	0.2193
4.320	73.1248	16.0301	0.2192
4.325	73.0921	16.0587	0.2197
4.330	73.1170	16.0621	0.2197
4.335	73.1304	16.1124	0.2203
4.340	73.1139	16.1135	0.2204
4.345	73.1172	16.1611	0.2210
4.350	73.1051	16.1605	0.2211
4.355	73.0962	16.1907	0.2215
4.360	73.0939	16.1847	0.2214
4.365	73.0888	16.1968	0.2216
4.370	73.0506	16.2093	0.2219
4.375	73.0938	16.2591	0.2224
4.380	73.0470	16.2516	0.2225
4.385	73.0393	16.2720	0.2228
4.390	73.0407	16.2860	0.2230
4.395	73.0654	16.3032	0.2231
4.400	73.0332	16.3229	0.2235
4.405	72.9814	16.3174	0.2236
4.410	73.0181	16.3548	0.2240
4.415	73.0135	16.3831	0.2244
4.420	72.9787	16.3858	0.2245
4.425	72.9616	16.3451	0.2240
4.430	72.9747	16.3994	0.2247
4.435	72.9631	16.4047	0.2248
4.440	72.9334	16.3933	0.2248
4.445	72.9505	16.4268	0.2252
4.450	72.9066	16.4168	0.2252
4.455	72.9089	16.4600	0.2258

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.460	72.9726	16.4283	0.2251
4.465	72.9446	16.4539	0.2256
4.470	72.9358	16.4786	0.2259
4.475	72.9409	16.5018	0.2262
4.480	72.9112	16.5081	0.2264
4.485	72.9047	16.4937	0.2262
4.490	72.9330	16.5177	0.2265
4.495	72.9363	16.5568	0.2270
4.500	72.8838	16.5135	0.2266
4.505	72.8745	16.5785	0.2275
4.510	72.8282	16.5665	0.2275
4.515	72.8476	16.5772	0.2276
4.520	72.7968	16.6331	0.2285
4.525	72.8369	16.5794	0.2276
4.530	72.7207	16.6142	0.2285
4.535	72.7631	16.6378	0.2287
4.540	72.7555	16.6557	0.2289
4.545	72.7613	16.6446	0.2288
4.550	72.7331	16.6578	0.2290
4.555	72.6708	16.6768	0.2295
4.560	72.6951	16.6880	0.2296
4.565	72.6803	16.6829	0.2295
4.570	72.6948	16.6609	0.2292
4.575	72.6414	16.6537	0.2293
4.580	72.6503	16.7169	0.2301
4.585	72.6420	16.6970	0.2299
4.590	72.6092	16.7606	0.2308
4.595	72.6322	16.8011	0.2313
4.600	72.5800	16.8258	0.2318
4.605	72.5878	16.8151	0.2317
4.610	72.5843	16.8256	0.2318
4.615	72.5076	16.7863	0.2315
4.620	72.5020	16.8764	0.2328
4.625	72.4462	16.8832	0.2330
4.630	72.4841	16.8871	0.2330
4.635	72.4749	16.8577	0.2326
4.640	72.4604	16.8544	0.2326
4.645	72.4740	16.8991	0.2332
4.650	72.4511	16.9162	0.2335
4.655	72.3879	16.9039	0.2335
4.660	72.3977	16.8950	0.2334
4.665	72.4124	16.8410	0.2326
4.670	72.3469	16.9183	0.2338
4.675	72.3430	16.9393	0.2342
4.680	72.3194	16.9025	0.2337
4.685	72.3416	16.9617	0.2345

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.690	72.3165	16.9931	0.2350
4.695	72.3040	16.9185	0.2340
4.700	72.3040	16.9566	0.2345
4.705	72.2404	16.9624	0.2348
4.710	72.3080	17.0134	0.2353
4.715	72.2975	17.0193	0.2354
4.720	72.3269	17.0112	0.2352
4.725	72.3761	17.0838	0.2360
4.730	72.2853	17.0982	0.2365
4.735	72.3246	17.1586	0.2372
4.740	72.3024	17.2147	0.2381
4.745	72.3286	17.2823	0.2389
4.750	72.3293	17.1729	0.2374
4.755	72.3094	17.2761	0.2389
4.760	72.3167	17.2830	0.2390
4.765	72.2441	17.3318	0.2399
4.770	72.2779	17.3484	0.2400
4.775	72.2925	17.3753	0.2403
4.780	72.2708	17.3249	0.2397
4.785	72.2221	17.4006	0.2409
4.790	72.3014	17.4363	0.2412
4.795	72.2559	17.4535	0.2416
4.800	72.2373	17.4324	0.2413
4.805	72.2528	17.4677	0.2418
4.810	72.2373	17.5149	0.2425
4.815	72.2219	17.5606	0.2431
4.820	72.2085	17.5090	0.2425
4.825	72.1301	17.5988	0.2440
4.830	72.1733	17.6292	0.2443
4.835	72.1757	17.6263	0.2442
4.840	72.1400	17.6337	0.2444
4.845	72.1586	17.6803	0.2450
4.850	72.1961	17.6428	0.2444
4.855	72.1673	17.6703	0.2449
4.860	72.1602	17.7375	0.2458
4.865	72.1060	17.6208	0.2444
4.870	72.1636	17.7075	0.2454
4.875	72.1589	17.7614	0.2461
4.880	72.1434	17.7961	0.2467
4.885	72.0860	17.8017	0.2470
4.890	72.1421	17.8500	0.2474
4.895	72.1120	17.8361	0.2473
4.900	72.0706	17.8624	0.2478
4.905	72.1091	17.8974	0.2482
4.910	72.0632	17.9039	0.2484
4.915	72.0714	17.9258	0.2487

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.920	72.0613	17.9398	0.2490
4.925	72.0450	17.9257	0.2488
4.930	72.0474	17.9016	0.2485
4.935	72.0498	17.9305	0.2489
4.940	72.0499	17.9417	0.2490
4.945	72.0527	17.9530	0.2492
4.950	72.1151	17.9534	0.2490
4.955	72.1069	17.9837	0.2494
4.960	72.0942	18.0093	0.2498
4.965	72.1333	18.0436	0.2501
4.970	72.1570	18.0811	0.2506
4.975	72.1258	18.0934	0.2509
4.980	72.1403	18.1401	0.2515
4.985	72.1400	18.1607	0.2517
4.990	72.1282	18.2296	0.2527
4.995	72.1014	18.2514	0.2531
5.000	72.0867	18.2678	0.2534
5.005	72.0670	18.2777	0.2536
5.010	72.0344	18.3003	0.2541
5.015	72.0317	18.2943	0.2540
5.020	72.0721	18.3132	0.2541
5.025	72.0290	18.3393	0.2546
5.030	72.0278	18.3545	0.2548
5.035	72.0357	18.4356	0.2559
5.040	72.0207	18.4670	0.2564
5.045	71.9737	18.4668	0.2566
5.050	71.9655	18.4857	0.2569
5.055	71.9398	18.4832	0.2569
5.060	71.8590	18.5699	0.2584
5.065	71.7848	18.5274	0.2581
5.070	71.7530	18.5056	0.2579
5.075	71.7213	18.4674	0.2575
5.080	71.6959	18.4818	0.2578
5.085	71.6733	18.4599	0.2576
5.090	71.6676	18.4354	0.2572
5.095	71.6998	18.4824	0.2578
5.100	71.6750	18.4293	0.2571
5.105	71.6522	18.4872	0.2580
5.110	71.6706	18.4395	0.2573
5.115	71.6631	18.4141	0.2570
5.120	71.6910	18.4258	0.2570
5.125	71.6919	18.4005	0.2567
5.130	71.7259	18.4463	0.2572
5.135	71.7490	18.3905	0.2563
5.140	71.7018	18.4471	0.2573
5.145	71.6903	18.5129	0.2582

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.150	71.7402	18.4998	0.2579
5.155	71.6997	18.4883	0.2579
5.160	71.6852	18.4611	0.2575
5.165	71.6867	18.4850	0.2579
5.170	71.6965	18.5485	0.2587
5.175	71.6349	18.5655	0.2592
5.180	71.6641	18.5359	0.2586
5.185	71.7216	18.5876	0.2592
5.190	71.6239	18.6023	0.2597
5.195	71.6528	18.5968	0.2595
5.200	71.6158	18.5723	0.2593
5.205	71.6187	18.6119	0.2599
5.210	71.5872	18.5893	0.2597
5.215	71.5489	18.6240	0.2603
5.220	71.5705	18.6306	0.2603
5.225	71.4896	18.6403	0.2607
5.230	71.5155	18.6473	0.2607
5.235	71.4980	18.6676	0.2611
5.240	71.4704	18.6783	0.2613
5.245	71.4556	18.7253	0.2621
5.250	71.4445	18.7204	0.2620
5.255	71.3970	18.7325	0.2624
5.260	71.4050	18.7488	0.2626
5.265	71.4041	18.7531	0.2626
5.270	71.3907	18.7568	0.2627
5.275	71.3454	18.7715	0.2631
5.280	71.3734	18.7766	0.2631
5.285	71.3353	18.7445	0.2628
5.290	71.3066	18.8057	0.2637
5.295	71.3060	18.8010	0.2637
5.300	71.3092	18.8064	0.2637
5.305	71.2574	18.7836	0.2636
5.310	71.2757	18.8012	0.2638
5.315	71.2884	18.8510	0.2644
5.320	71.2359	18.8405	0.2645
5.325	71.2314	18.8579	0.2647
5.330	71.2245	18.8693	0.2649
5.335	71.2375	18.8851	0.2651
5.340	71.1926	18.8920	0.2654
5.345	71.1080	18.9208	0.2661
5.350	71.2011	18.9141	0.2656
5.355	71.1624	18.9332	0.2661
5.360	71.1576	18.9332	0.2661
5.365	71.1302	19.0289	0.2675
5.370	71.1339	19.0218	0.2674
5.375	71.1095	18.9858	0.2670

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.380	71.1141	19.0433	0.2678
5.385	71.1403	19.0328	0.2675
5.390	71.1514	19.1056	0.2685
5.395	71.1802	19.0924	0.2682
5.400	71.1352	19.0979	0.2685
5.405	71.1254	19.1442	0.2692
5.410	71.1148	19.1530	0.2693
5.415	71.1421	19.1991	0.2699
5.420	71.0983	19.1635	0.2695
5.425	71.0503	19.2700	0.2712
5.430	71.0750	19.2385	0.2707
5.435	71.0942	19.2957	0.2714
5.440	71.0869	19.3551	0.2723
5.445	71.0862	19.3518	0.2722
5.450	71.0763	19.3130	0.2717
5.455	71.0456	19.3136	0.2718
5.460	71.0731	19.3817	0.2727
5.465	71.1185	19.3910	0.2727
5.470	71.0445	19.4572	0.2739
5.475	71.0903	19.4628	0.2738
5.480	71.0339	19.5192	0.2748
5.485	71.0101	19.5304	0.2750
5.490	71.0171	19.5557	0.2754
5.495	71.0669	19.5586	0.2752
5.500	71.0806	19.6022	0.2758
5.505	71.1047	19.5653	0.2752
5.510	71.0159	19.6441	0.2766
5.515	71.0183	19.6432	0.2766
5.520	70.9986	19.6744	0.2771
5.525	71.0023	19.6941	0.2774
5.530	71.0235	19.6941	0.2773
5.535	71.0375	19.7078	0.2774
5.540	70.9781	19.7450	0.2782
5.545	71.0191	19.7477	0.2781
5.550	70.9966	19.7751	0.2785
5.555	70.9828	19.7895	0.2788
5.560	70.9877	19.8359	0.2794
5.565	70.9732	19.8321	0.2794
5.570	70.9626	19.8594	0.2799
5.575	70.9747	19.8838	0.2802
5.580	70.9768	19.9079	0.2805
5.585	70.9738	19.9140	0.2806
5.590	70.9554	19.9303	0.2809
5.595	70.9528	19.9460	0.2811
5.600	70.9756	19.9766	0.2815
5.605	70.9649	19.9832	0.2816

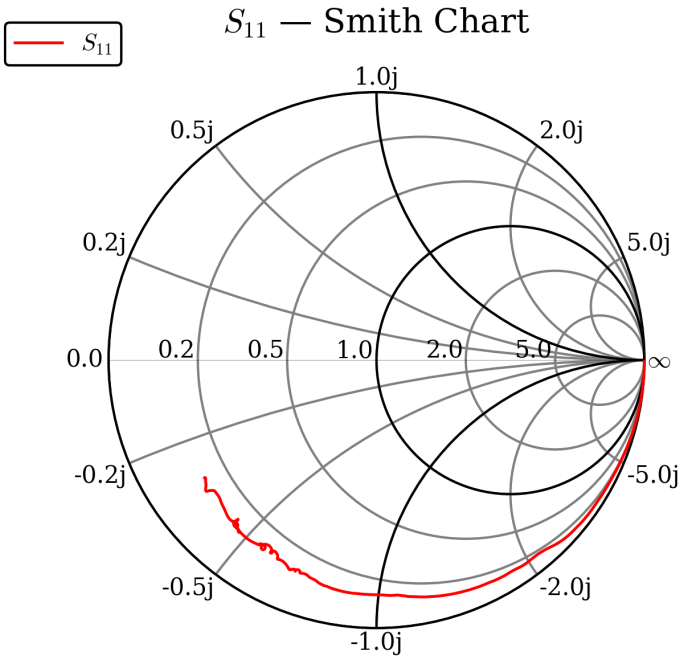
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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.610	70.9042	19.9863	0.2819
5.615	70.9285	19.9970	0.2819
5.620	70.9091	19.9927	0.2819
5.625	70.9360	20.0270	0.2823
5.630	70.8888	20.0311	0.2826
5.635	70.9063	20.0329	0.2825
5.640	70.9062	20.0850	0.2833
5.645	70.8847	20.0596	0.2830
5.650	70.8699	20.0879	0.2834
5.655	70.8412	20.0659	0.2833
5.660	70.8437	20.0546	0.2831
5.665	70.8703	20.1073	0.2837
5.670	70.8245	20.1517	0.2845
5.675	70.8293	20.1387	0.2843
5.680	70.7372	20.1108	0.2843
5.685	70.8071	20.0779	0.2836
5.690	70.7647	20.1204	0.2843
5.695	70.7887	20.1098	0.2841
5.700	70.8088	20.1389	0.2844
5.705	70.7258	20.2176	0.2859
5.710	70.7770	20.1030	0.2840
5.715	70.7483	20.1911	0.2854
5.720	70.7544	20.2285	0.2859
5.725	70.8016	20.2059	0.2854
5.730	70.7176	20.1635	0.2851
5.735	70.7567	20.2257	0.2858
5.740	70.7099	20.2165	0.2859
5.745	70.6952	20.3704	0.2881
5.750	70.6999	20.2664	0.2867
5.755	70.6930	20.2705	0.2867
5.760	70.7210	20.2963	0.2870
5.765	70.6165	20.2624	0.2869
5.770	70.6903	20.3461	0.2878
5.775	70.5959	20.2938	0.2875
5.780	70.7083	20.3239	0.2874
5.785	70.5795	20.3343	0.2881
5.790	70.6338	20.4101	0.2890
5.795	70.6004	20.3894	0.2888
5.800	70.5364	20.3359	0.2883
5.805	70.5427	20.3906	0.2891
5.810	70.5279	20.3861	0.2890
5.815	70.5178	20.3904	0.2892
5.820	70.4671	20.3728	0.2891
5.825	70.4836	20.4550	0.2902
5.830	70.4232	20.4128	0.2899
5.835	70.4152	20.4162	0.2899

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.840	70.4153	20.3946	0.2896
5.845	70.3279	20.4198	0.2904
5.850	70.3249	20.5053	0.2916
5.855	70.3625	20.4753	0.2910
5.860	70.3151	20.4905	0.2914
5.865	70.3393	20.5337	0.2919
5.870	70.2915	20.5299	0.2921
5.875	70.2971	20.5517	0.2924
5.880	70.2693	20.5401	0.2923
5.885	70.2625	20.5766	0.2929
5.890	70.2667	20.5672	0.2927
5.895	70.2776	20.5159	0.2919
5.900	70.2455	20.5593	0.2927
5.905	70.2675	20.6110	0.2933
5.910	70.2252	20.6086	0.2935
5.915	70.2279	20.5817	0.2931
5.920	70.1836	20.5783	0.2932
5.925	70.1565	20.6228	0.2940
5.930	70.1345	20.6279	0.2941
5.935	70.0833	20.6187	0.2942
5.940	70.1039	20.6371	0.2944
5.945	70.1550	20.6532	0.2944
5.950	70.1191	20.7061	0.2953
5.955	70.1120	20.7192	0.2955
5.960	70.0792	20.7448	0.2960
5.965	70.1005	20.8271	0.2971
5.970	70.0371	20.7805	0.2967
5.975	69.9440	20.8071	0.2975
5.980	70.1024	20.8228	0.2970
5.985	70.0129	20.8611	0.2980
5.990	69.9825	20.8504	0.2979
5.995	70.0068	20.9208	0.2988
6.000	70.0201	20.8635	0.2980

1.2 S_{11} — Smith Chart



Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
0.005	0.9997	-0.0050	-0.00
0.010	0.9997	-0.0098	-0.00
0.015	0.9995	-0.0148	-0.00
0.020	0.9993	-0.0198	-0.00
0.025	0.9993	-0.0247	-0.00
0.030	0.9991	-0.0297	-0.00
0.035	0.9990	-0.0344	-0.00
0.040	0.9987	-0.0394	-0.00
0.045	0.9986	-0.0444	-0.00
0.050	0.9982	-0.0494	-0.00
0.055	0.9980	-0.0543	-0.00
0.060	0.9975	-0.0592	-0.01
0.065	0.9972	-0.0641	-0.01
0.070	0.9968	-0.0689	-0.01
0.075	0.9964	-0.0738	-0.01
0.080	0.9960	-0.0786	-0.01
0.085	0.9955	-0.0834	-0.01
0.090	0.9950	-0.0884	-0.01
0.095	0.9944	-0.0933	-0.01
0.100	0.9939	-0.0982	-0.01
0.105	0.9932	-0.1030	-0.01
0.110	0.9926	-0.1079	-0.01
0.115	0.9920	-0.1126	-0.01

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.120	0.9913	-0.1175	-0.02
0.125	0.9906	-0.1223	-0.02
0.130	0.9900	-0.1271	-0.02
0.135	0.9893	-0.1319	-0.02
0.140	0.9885	-0.1367	-0.02
0.145	0.9877	-0.1416	-0.02
0.150	0.9868	-0.1464	-0.02
0.155	0.9861	-0.1512	-0.02
0.160	0.9851	-0.1558	-0.02
0.165	0.9843	-0.1607	-0.02
0.170	0.9834	-0.1654	-0.02
0.175	0.9824	-0.1702	-0.03
0.180	0.9816	-0.1749	-0.03
0.185	0.9804	-0.1798	-0.03
0.190	0.9792	-0.1845	-0.03
0.195	0.9782	-0.1892	-0.03
0.200	0.9772	-0.1939	-0.03
0.205	0.9761	-0.1987	-0.03
0.210	0.9749	-0.2033	-0.04
0.215	0.9738	-0.2080	-0.04
0.220	0.9726	-0.2127	-0.04
0.225	0.9713	-0.2176	-0.04
0.230	0.9701	-0.2222	-0.04
0.235	0.9688	-0.2269	-0.04
0.240	0.9675	-0.2315	-0.04
0.245	0.9661	-0.2362	-0.05
0.250	0.9647	-0.2409	-0.05
0.255	0.9634	-0.2454	-0.05
0.260	0.9620	-0.2502	-0.05
0.265	0.9606	-0.2548	-0.05
0.270	0.9592	-0.2593	-0.06
0.275	0.9577	-0.2639	-0.06
0.280	0.9561	-0.2685	-0.06
0.285	0.9546	-0.2730	-0.06
0.290	0.9529	-0.2776	-0.07
0.295	0.9515	-0.2821	-0.07
0.300	0.9498	-0.2865	-0.07
0.305	0.9482	-0.2912	-0.07
0.310	0.9465	-0.2957	-0.07
0.315	0.9449	-0.3000	-0.07
0.320	0.9432	-0.3045	-0.08
0.325	0.9415	-0.3089	-0.08
0.330	0.9397	-0.3134	-0.08
0.335	0.9381	-0.3176	-0.08
0.340	0.9362	-0.3221	-0.09
0.345	0.9344	-0.3266	-0.09

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.350	0.9326	-0.3309	-0.09
0.355	0.9307	-0.3352	-0.09
0.360	0.9289	-0.3396	-0.10
0.365	0.9269	-0.3439	-0.10
0.370	0.9250	-0.3482	-0.10
0.375	0.9230	-0.3525	-0.11
0.380	0.9210	-0.3567	-0.11
0.385	0.9192	-0.3610	-0.11
0.390	0.9172	-0.3653	-0.11
0.395	0.9153	-0.3694	-0.11
0.400	0.9135	-0.3735	-0.11
0.405	0.9116	-0.3777	-0.12
0.410	0.9094	-0.3817	-0.12
0.415	0.9075	-0.3859	-0.12
0.420	0.9056	-0.3900	-0.12
0.425	0.9034	-0.3940	-0.13
0.430	0.9014	-0.3981	-0.13
0.435	0.8995	-0.4021	-0.13
0.440	0.8973	-0.4062	-0.13
0.445	0.8953	-0.4102	-0.13
0.450	0.8931	-0.4140	-0.14
0.455	0.8910	-0.4180	-0.14
0.460	0.8889	-0.4219	-0.14
0.465	0.8867	-0.4257	-0.14
0.470	0.8846	-0.4297	-0.14
0.475	0.8824	-0.4335	-0.15
0.480	0.8803	-0.4374	-0.15
0.485	0.8782	-0.4412	-0.15
0.490	0.8761	-0.4450	-0.15
0.495	0.8739	-0.4488	-0.15
0.500	0.8718	-0.4525	-0.16
0.505	0.8697	-0.4564	-0.16
0.510	0.8674	-0.4601	-0.16
0.515	0.8653	-0.4638	-0.16
0.520	0.8631	-0.4675	-0.16
0.525	0.8611	-0.4713	-0.16
0.530	0.8589	-0.4747	-0.16
0.535	0.8569	-0.4785	-0.16
0.540	0.8546	-0.4822	-0.16
0.545	0.8523	-0.4857	-0.17
0.550	0.8500	-0.4895	-0.17
0.555	0.8477	-0.4930	-0.17
0.560	0.8455	-0.4967	-0.17
0.565	0.8433	-0.5002	-0.17
0.570	0.8409	-0.5037	-0.17
0.575	0.8386	-0.5075	-0.17

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.580	0.8361	-0.5110	-0.18
0.585	0.8338	-0.5145	-0.18
0.590	0.8313	-0.5177	-0.18
0.595	0.8291	-0.5211	-0.18
0.600	0.8268	-0.5244	-0.18
0.605	0.8245	-0.5278	-0.18
0.610	0.8222	-0.5311	-0.19
0.615	0.8200	-0.5344	-0.19
0.620	0.8177	-0.5379	-0.19
0.625	0.8153	-0.5413	-0.19
0.630	0.8130	-0.5447	-0.19
0.635	0.8106	-0.5482	-0.19
0.640	0.8083	-0.5515	-0.19
0.645	0.8059	-0.5548	-0.19
0.650	0.8035	-0.5582	-0.19
0.655	0.8011	-0.5614	-0.19
0.660	0.7986	-0.5648	-0.19
0.665	0.7961	-0.5680	-0.19
0.670	0.7937	-0.5714	-0.19
0.675	0.7912	-0.5746	-0.19
0.680	0.7887	-0.5778	-0.20
0.685	0.7861	-0.5809	-0.20
0.690	0.7835	-0.5841	-0.20
0.695	0.7808	-0.5874	-0.20
0.700	0.7782	-0.5905	-0.20
0.705	0.7755	-0.5939	-0.20
0.710	0.7727	-0.5972	-0.21
0.715	0.7700	-0.6003	-0.21
0.720	0.7671	-0.6034	-0.21
0.725	0.7642	-0.6065	-0.21
0.730	0.7613	-0.6095	-0.22
0.735	0.7585	-0.6126	-0.22
0.740	0.7556	-0.6154	-0.22
0.745	0.7526	-0.6182	-0.23
0.750	0.7498	-0.6211	-0.23
0.755	0.7470	-0.6239	-0.23
0.760	0.7444	-0.6267	-0.24
0.765	0.7414	-0.6296	-0.24
0.770	0.7385	-0.6326	-0.24
0.775	0.7357	-0.6356	-0.24
0.780	0.7327	-0.6388	-0.25
0.785	0.7296	-0.6419	-0.25
0.790	0.7266	-0.6448	-0.25
0.795	0.7233	-0.6479	-0.26
0.800	0.7200	-0.6508	-0.26
0.805	0.7169	-0.6538	-0.26

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.810	0.7135	-0.6565	-0.27
0.815	0.7100	-0.6594	-0.27
0.820	0.7066	-0.6621	-0.28
0.825	0.7033	-0.6648	-0.28
0.830	0.6996	-0.6676	-0.29
0.835	0.6962	-0.6702	-0.30
0.840	0.6924	-0.6727	-0.31
0.845	0.6889	-0.6752	-0.31
0.850	0.6854	-0.6776	-0.32
0.855	0.6816	-0.6801	-0.33
0.860	0.6780	-0.6823	-0.34
0.865	0.6742	-0.6845	-0.35
0.870	0.6706	-0.6865	-0.36
0.875	0.6669	-0.6885	-0.37
0.880	0.6633	-0.6905	-0.38
0.885	0.6597	-0.6923	-0.39
0.890	0.6562	-0.6941	-0.40
0.895	0.6526	-0.6960	-0.41
0.900	0.6493	-0.6975	-0.42
0.905	0.6459	-0.6992	-0.43
0.910	0.6426	-0.7007	-0.44
0.915	0.6394	-0.7023	-0.45
0.920	0.6363	-0.7039	-0.46
0.925	0.6332	-0.7054	-0.46
0.930	0.6303	-0.7071	-0.47
0.935	0.6273	-0.7090	-0.48
0.940	0.6243	-0.7107	-0.48
0.945	0.6213	-0.7124	-0.49
0.950	0.6182	-0.7144	-0.49
0.955	0.6152	-0.7160	-0.50
0.960	0.6124	-0.7178	-0.51
0.965	0.6094	-0.7194	-0.51
0.970	0.6068	-0.7212	-0.51
0.975	0.6041	-0.7228	-0.52
0.980	0.6015	-0.7245	-0.52
0.985	0.5990	-0.7264	-0.52
0.990	0.5964	-0.7283	-0.53
0.995	0.5940	-0.7302	-0.53
1.000	0.5914	-0.7324	-0.52
1.005	0.5888	-0.7345	-0.52
1.010	0.5861	-0.7366	-0.52
1.015	0.5833	-0.7388	-0.53
1.020	0.5807	-0.7410	-0.52
1.025	0.5778	-0.7435	-0.52
1.030	0.5749	-0.7457	-0.52
1.035	0.5721	-0.7480	-0.52

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.040	0.5691	-0.7503	-0.52
1.045	0.5662	-0.7526	-0.52
1.050	0.5632	-0.7548	-0.52
1.055	0.5602	-0.7572	-0.52
1.060	0.5570	-0.7593	-0.52
1.065	0.5538	-0.7615	-0.52
1.070	0.5507	-0.7638	-0.52
1.075	0.5474	-0.7658	-0.53
1.080	0.5441	-0.7677	-0.53
1.085	0.5409	-0.7696	-0.53
1.090	0.5375	-0.7715	-0.54
1.095	0.5342	-0.7732	-0.54
1.100	0.5308	-0.7748	-0.55
1.105	0.5277	-0.7761	-0.55
1.110	0.5246	-0.7777	-0.56
1.115	0.5215	-0.7792	-0.56
1.120	0.5186	-0.7804	-0.56
1.125	0.5158	-0.7818	-0.57
1.130	0.5131	-0.7833	-0.57
1.135	0.5102	-0.7848	-0.57
1.140	0.5076	-0.7863	-0.58
1.145	0.5049	-0.7877	-0.58
1.150	0.5024	-0.7893	-0.58
1.155	0.4996	-0.7911	-0.58
1.160	0.4970	-0.7926	-0.58
1.165	0.4943	-0.7942	-0.58
1.170	0.4918	-0.7959	-0.58
1.175	0.4890	-0.7975	-0.58
1.180	0.4865	-0.7991	-0.58
1.185	0.4839	-0.8007	-0.58
1.190	0.4812	-0.8023	-0.58
1.195	0.4787	-0.8037	-0.58
1.200	0.4760	-0.8053	-0.58
1.205	0.4733	-0.8066	-0.58
1.210	0.4707	-0.8080	-0.58
1.215	0.4679	-0.8095	-0.58
1.220	0.4651	-0.8108	-0.59
1.225	0.4624	-0.8122	-0.59
1.230	0.4596	-0.8136	-0.59
1.235	0.4569	-0.8150	-0.59
1.240	0.4541	-0.8164	-0.59
1.245	0.4514	-0.8176	-0.59
1.250	0.4486	-0.8187	-0.60
1.255	0.4458	-0.8200	-0.60
1.260	0.4432	-0.8212	-0.60
1.265	0.4407	-0.8222	-0.60

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.270	0.4380	-0.8233	-0.61
1.275	0.4353	-0.8242	-0.61
1.280	0.4327	-0.8252	-0.61
1.285	0.4299	-0.8262	-0.62
1.290	0.4274	-0.8273	-0.62
1.295	0.4246	-0.8283	-0.62
1.300	0.4222	-0.8294	-0.62
1.305	0.4197	-0.8303	-0.63
1.310	0.4172	-0.8314	-0.63
1.315	0.4150	-0.8324	-0.63
1.320	0.4125	-0.8335	-0.63
1.325	0.4100	-0.8346	-0.63
1.330	0.4076	-0.8354	-0.63
1.335	0.4051	-0.8367	-0.63
1.340	0.4027	-0.8378	-0.63
1.345	0.4000	-0.8387	-0.64
1.350	0.3975	-0.8397	-0.64
1.355	0.3947	-0.8408	-0.64
1.360	0.3922	-0.8418	-0.64
1.365	0.3897	-0.8429	-0.64
1.370	0.3870	-0.8440	-0.64
1.375	0.3843	-0.8449	-0.65
1.380	0.3815	-0.8457	-0.65
1.385	0.3788	-0.8468	-0.65
1.390	0.3760	-0.8476	-0.66
1.395	0.3731	-0.8488	-0.66
1.400	0.3702	-0.8496	-0.66
1.405	0.3673	-0.8507	-0.66
1.410	0.3643	-0.8515	-0.67
1.415	0.3613	-0.8525	-0.67
1.420	0.3584	-0.8536	-0.67
1.425	0.3553	-0.8546	-0.67
1.430	0.3521	-0.8554	-0.68
1.435	0.3491	-0.8565	-0.68
1.440	0.3458	-0.8576	-0.68
1.445	0.3427	-0.8584	-0.68
1.450	0.3396	-0.8592	-0.69
1.455	0.3365	-0.8602	-0.69
1.460	0.3334	-0.8611	-0.69
1.465	0.3300	-0.8617	-0.70
1.470	0.3269	-0.8627	-0.70
1.475	0.3237	-0.8635	-0.70
1.480	0.3203	-0.8642	-0.71
1.485	0.3169	-0.8651	-0.71
1.490	0.3137	-0.8660	-0.71
1.495	0.3103	-0.8668	-0.72

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.500	0.3070	-0.8678	-0.72
1.505	0.3035	-0.8684	-0.72
1.510	0.3001	-0.8690	-0.73
1.515	0.2967	-0.8700	-0.73
1.520	0.2931	-0.8708	-0.74
1.525	0.2898	-0.8714	-0.74
1.530	0.2860	-0.8723	-0.74
1.535	0.2823	-0.8730	-0.75
1.540	0.2787	-0.8735	-0.75
1.545	0.2750	-0.8743	-0.76
1.550	0.2714	-0.8748	-0.76
1.555	0.2676	-0.8752	-0.77
1.560	0.2637	-0.8759	-0.77
1.565	0.2599	-0.8764	-0.78
1.570	0.2562	-0.8771	-0.78
1.575	0.2526	-0.8777	-0.79
1.580	0.2488	-0.8781	-0.79
1.585	0.2451	-0.8786	-0.80
1.590	0.2414	-0.8792	-0.80
1.595	0.2377	-0.8798	-0.81
1.600	0.2341	-0.8802	-0.81
1.605	0.2303	-0.8806	-0.82
1.610	0.2268	-0.8807	-0.82
1.615	0.2231	-0.8811	-0.83
1.620	0.2194	-0.8813	-0.84
1.625	0.2158	-0.8817	-0.84
1.630	0.2120	-0.8820	-0.85
1.635	0.2083	-0.8822	-0.85
1.640	0.2046	-0.8824	-0.86
1.645	0.2010	-0.8826	-0.87
1.650	0.1974	-0.8827	-0.87
1.655	0.1938	-0.8829	-0.88
1.660	0.1902	-0.8832	-0.88
1.665	0.1866	-0.8835	-0.89
1.670	0.1831	-0.8837	-0.89
1.675	0.1794	-0.8836	-0.90
1.680	0.1760	-0.8837	-0.91
1.685	0.1725	-0.8837	-0.91
1.690	0.1692	-0.8837	-0.92
1.695	0.1659	-0.8837	-0.92
1.700	0.1625	-0.8836	-0.93
1.705	0.1592	-0.8834	-0.94
1.710	0.1556	-0.8836	-0.94
1.715	0.1524	-0.8833	-0.95
1.720	0.1490	-0.8831	-0.96
1.725	0.1458	-0.8829	-0.96

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.730	0.1424	-0.8829	-0.97
1.735	0.1394	-0.8827	-0.98
1.740	0.1363	-0.8826	-0.98
1.745	0.1333	-0.8823	-0.99
1.750	0.1303	-0.8826	-0.99
1.755	0.1276	-0.8825	-1.00
1.760	0.1247	-0.8823	-1.00
1.765	0.1220	-0.8822	-1.01
1.770	0.1194	-0.8820	-1.01
1.775	0.1169	-0.8818	-1.02
1.780	0.1147	-0.8813	-1.02
1.785	0.1119	-0.8810	-1.03
1.790	0.1093	-0.8806	-1.04
1.795	0.1067	-0.8800	-1.05
1.800	0.1042	-0.8797	-1.05
1.805	0.1018	-0.8795	-1.06
1.810	0.0996	-0.8793	-1.06
1.815	0.0974	-0.8792	-1.06
1.820	0.0954	-0.8788	-1.07
1.825	0.0936	-0.8786	-1.08
1.830	0.0915	-0.8782	-1.08
1.835	0.0895	-0.8780	-1.09
1.840	0.0877	-0.8776	-1.09
1.845	0.0858	-0.8773	-1.10
1.850	0.0839	-0.8771	-1.10
1.855	0.0821	-0.8770	-1.10
1.860	0.0802	-0.8766	-1.11
1.865	0.0783	-0.8765	-1.11
1.870	0.0765	-0.8765	-1.11
1.875	0.0745	-0.8767	-1.11
1.880	0.0728	-0.8771	-1.11
1.885	0.0712	-0.8775	-1.11
1.890	0.0697	-0.8775	-1.11
1.895	0.0682	-0.8775	-1.11
1.900	0.0664	-0.8775	-1.11
1.905	0.0645	-0.8774	-1.11
1.910	0.0628	-0.8775	-1.11
1.915	0.0610	-0.8776	-1.11
1.920	0.0591	-0.8779	-1.11
1.925	0.0574	-0.8782	-1.11
1.930	0.0558	-0.8782	-1.11
1.935	0.0542	-0.8782	-1.11
1.940	0.0524	-0.8783	-1.11
1.945	0.0507	-0.8781	-1.11
1.950	0.0491	-0.8782	-1.11
1.955	0.0475	-0.8781	-1.12

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.960	0.0459	-0.8781	-1.12
1.965	0.0442	-0.8780	-1.12
1.970	0.0428	-0.8778	-1.12
1.975	0.0414	-0.8776	-1.12
1.980	0.0398	-0.8776	-1.12
1.985	0.0382	-0.8775	-1.13
1.990	0.0366	-0.8772	-1.13
1.995	0.0352	-0.8772	-1.13
2.000	0.0334	-0.8770	-1.13
2.005	0.0320	-0.8769	-1.14
2.010	0.0304	-0.8769	-1.14
2.015	0.0288	-0.8768	-1.14
2.020	0.0273	-0.8765	-1.14
2.025	0.0258	-0.8766	-1.14
2.030	0.0241	-0.8767	-1.14
2.035	0.0226	-0.8764	-1.14
2.040	0.0210	-0.8763	-1.14
2.045	0.0192	-0.8762	-1.15
2.050	0.0177	-0.8762	-1.15
2.055	0.0159	-0.8762	-1.15
2.060	0.0142	-0.8760	-1.15
2.065	0.0123	-0.8762	-1.15
2.070	0.0105	-0.8760	-1.15
2.075	0.0086	-0.8758	-1.15
2.080	0.0067	-0.8759	-1.15
2.085	0.0047	-0.8756	-1.15
2.090	0.0026	-0.8755	-1.15
2.095	0.0004	-0.8753	-1.16
2.100	-0.0017	-0.8755	-1.15
2.105	-0.0039	-0.8752	-1.16
2.110	-0.0063	-0.8753	-1.16
2.115	-0.0086	-0.8752	-1.16
2.120	-0.0109	-0.8753	-1.16
2.125	-0.0133	-0.8753	-1.16
2.130	-0.0158	-0.8751	-1.16
2.135	-0.0181	-0.8751	-1.16
2.140	-0.0206	-0.8749	-1.16
2.145	-0.0230	-0.8748	-1.16
2.150	-0.0257	-0.8749	-1.16
2.155	-0.0282	-0.8747	-1.16
2.160	-0.0306	-0.8746	-1.16
2.165	-0.0334	-0.8745	-1.16
2.170	-0.0360	-0.8743	-1.16
2.175	-0.0387	-0.8740	-1.16
2.180	-0.0413	-0.8739	-1.16
2.185	-0.0440	-0.8737	-1.16

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.190	-0.0467	-0.8733	-1.16
2.195	-0.0495	-0.8734	-1.16
2.200	-0.0524	-0.8733	-1.16
2.205	-0.0552	-0.8729	-1.16
2.210	-0.0581	-0.8727	-1.16
2.215	-0.0608	-0.8723	-1.17
2.220	-0.0638	-0.8721	-1.17
2.225	-0.0667	-0.8718	-1.17
2.230	-0.0694	-0.8715	-1.17
2.235	-0.0724	-0.8712	-1.17
2.240	-0.0754	-0.8709	-1.17
2.245	-0.0782	-0.8707	-1.17
2.250	-0.0811	-0.8704	-1.17
2.255	-0.0841	-0.8701	-1.17
2.260	-0.0869	-0.8699	-1.17
2.265	-0.0899	-0.8697	-1.17
2.270	-0.0928	-0.8691	-1.17
2.275	-0.0956	-0.8689	-1.17
2.280	-0.0985	-0.8683	-1.17
2.285	-0.1013	-0.8680	-1.17
2.290	-0.1040	-0.8674	-1.17
2.295	-0.1069	-0.8668	-1.18
2.300	-0.1097	-0.8665	-1.18
2.305	-0.1125	-0.8659	-1.18
2.310	-0.1154	-0.8655	-1.18
2.315	-0.1181	-0.8650	-1.18
2.320	-0.1207	-0.8644	-1.18
2.325	-0.1236	-0.8639	-1.18
2.330	-0.1263	-0.8632	-1.19
2.335	-0.1287	-0.8626	-1.19
2.340	-0.1314	-0.8620	-1.19
2.345	-0.1340	-0.8614	-1.19
2.350	-0.1364	-0.8607	-1.20
2.355	-0.1389	-0.8600	-1.20
2.360	-0.1411	-0.8594	-1.20
2.365	-0.1434	-0.8587	-1.20
2.370	-0.1458	-0.8581	-1.21
2.375	-0.1479	-0.8574	-1.21
2.380	-0.1502	-0.8570	-1.21
2.385	-0.1523	-0.8563	-1.21
2.390	-0.1544	-0.8558	-1.21
2.395	-0.1564	-0.8552	-1.22
2.400	-0.1584	-0.8546	-1.22
2.405	-0.1603	-0.8541	-1.22
2.410	-0.1624	-0.8532	-1.22
2.415	-0.1642	-0.8527	-1.23

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.420	-0.1659	-0.8522	-1.23
2.425	-0.1677	-0.8515	-1.23
2.430	-0.1693	-0.8509	-1.23
2.435	-0.1709	-0.8502	-1.24
2.440	-0.1726	-0.8498	-1.24
2.445	-0.1739	-0.8490	-1.24
2.450	-0.1754	-0.8485	-1.25
2.455	-0.1765	-0.8479	-1.25
2.460	-0.1778	-0.8472	-1.25
2.465	-0.1791	-0.8466	-1.26
2.470	-0.1803	-0.8461	-1.26
2.475	-0.1816	-0.8457	-1.26
2.480	-0.1830	-0.8451	-1.26
2.485	-0.1840	-0.8446	-1.27
2.490	-0.1853	-0.8442	-1.27
2.495	-0.1866	-0.8439	-1.27
2.500	-0.1878	-0.8434	-1.27
2.505	-0.1888	-0.8431	-1.27
2.510	-0.1899	-0.8428	-1.27
2.515	-0.1910	-0.8425	-1.27
2.520	-0.1922	-0.8421	-1.27
2.525	-0.1933	-0.8418	-1.27
2.530	-0.1944	-0.8417	-1.27
2.535	-0.1955	-0.8412	-1.27
2.540	-0.1966	-0.8412	-1.27
2.545	-0.1975	-0.8409	-1.27
2.550	-0.1986	-0.8406	-1.27
2.555	-0.1994	-0.8401	-1.28
2.560	-0.2001	-0.8397	-1.28
2.565	-0.2006	-0.8392	-1.28
2.570	-0.2010	-0.8388	-1.28
2.575	-0.2015	-0.8383	-1.29
2.580	-0.2019	-0.8378	-1.29
2.585	-0.2023	-0.8377	-1.29
2.590	-0.2027	-0.8375	-1.29
2.595	-0.2033	-0.8374	-1.29
2.600	-0.2039	-0.8373	-1.29
2.605	-0.2043	-0.8373	-1.29
2.610	-0.2050	-0.8371	-1.29
2.615	-0.2056	-0.8369	-1.29
2.620	-0.2062	-0.8366	-1.29
2.625	-0.2066	-0.8364	-1.29
2.630	-0.2073	-0.8361	-1.30
2.635	-0.2079	-0.8357	-1.30
2.640	-0.2084	-0.8353	-1.30
2.645	-0.2090	-0.8349	-1.30

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.650	-0.2096	-0.8346	-1.31
2.655	-0.2103	-0.8343	-1.31
2.660	-0.2108	-0.8338	-1.31
2.665	-0.2113	-0.8332	-1.31
2.670	-0.2122	-0.8327	-1.32
2.675	-0.2127	-0.8326	-1.32
2.680	-0.2132	-0.8323	-1.32
2.685	-0.2137	-0.8321	-1.32
2.690	-0.2140	-0.8320	-1.32
2.695	-0.2147	-0.8318	-1.32
2.700	-0.2151	-0.8315	-1.32
2.705	-0.2156	-0.8315	-1.32
2.710	-0.2161	-0.8310	-1.32
2.715	-0.2165	-0.8306	-1.33
2.720	-0.2170	-0.8303	-1.33
2.725	-0.2177	-0.8298	-1.33
2.730	-0.2184	-0.8291	-1.34
2.735	-0.2192	-0.8283	-1.34
2.740	-0.2202	-0.8276	-1.35
2.745	-0.2210	-0.8270	-1.35
2.750	-0.2217	-0.8261	-1.36
2.755	-0.2229	-0.8252	-1.36
2.760	-0.2242	-0.8244	-1.37
2.765	-0.2252	-0.8235	-1.37
2.770	-0.2263	-0.8227	-1.38
2.775	-0.2276	-0.8217	-1.38
2.780	-0.2289	-0.8211	-1.39
2.785	-0.2303	-0.8204	-1.39
2.790	-0.2316	-0.8195	-1.40
2.795	-0.2330	-0.8188	-1.40
2.800	-0.2343	-0.8181	-1.40
2.805	-0.2356	-0.8175	-1.40
2.810	-0.2369	-0.8168	-1.41
2.815	-0.2384	-0.8161	-1.41
2.820	-0.2398	-0.8158	-1.41
2.825	-0.2410	-0.8149	-1.41
2.830	-0.2425	-0.8142	-1.42
2.835	-0.2439	-0.8137	-1.42
2.840	-0.2451	-0.8131	-1.42
2.845	-0.2463	-0.8124	-1.42
2.850	-0.2477	-0.8116	-1.43
2.855	-0.2488	-0.8110	-1.43
2.860	-0.2502	-0.8104	-1.43
2.865	-0.2511	-0.8098	-1.43
2.870	-0.2518	-0.8089	-1.44
2.875	-0.2529	-0.8080	-1.45

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.880	-0.2538	-0.8071	-1.45
2.885	-0.2548	-0.8058	-1.46
2.890	-0.2562	-0.8046	-1.47
2.895	-0.2575	-0.8032	-1.48
2.900	-0.2591	-0.8021	-1.48
2.905	-0.2605	-0.8010	-1.49
2.910	-0.2620	-0.7999	-1.50
2.915	-0.2639	-0.7989	-1.50
2.920	-0.2656	-0.7982	-1.50
2.925	-0.2671	-0.7977	-1.50
2.930	-0.2687	-0.7969	-1.50
2.935	-0.2702	-0.7965	-1.50
2.940	-0.2717	-0.7963	-1.50
2.945	-0.2730	-0.7960	-1.50
2.950	-0.2742	-0.7959	-1.50
2.955	-0.2755	-0.7959	-1.49
2.960	-0.2763	-0.7957	-1.49
2.965	-0.2772	-0.7957	-1.49
2.970	-0.2780	-0.7957	-1.49
2.975	-0.2787	-0.7956	-1.48
2.980	-0.2795	-0.7957	-1.48
2.985	-0.2798	-0.7955	-1.48
2.990	-0.2802	-0.7954	-1.48
2.995	-0.2804	-0.7952	-1.48
3.000	-0.2812	-0.7948	-1.48
3.005	-0.2842	-0.7936	-1.48
3.010	-0.2848	-0.7936	-1.48
3.015	-0.2851	-0.7937	-1.48
3.020	-0.2852	-0.7935	-1.48
3.025	-0.2854	-0.7940	-1.48
3.030	-0.2853	-0.7935	-1.48
3.035	-0.2853	-0.7930	-1.49
3.040	-0.2853	-0.7921	-1.49
3.045	-0.2852	-0.7916	-1.50
3.050	-0.2848	-0.7911	-1.51
3.055	-0.2850	-0.7899	-1.52
3.060	-0.2854	-0.7889	-1.53
3.065	-0.2860	-0.7877	-1.53
3.070	-0.2866	-0.7868	-1.54
3.075	-0.2874	-0.7858	-1.55
3.080	-0.2878	-0.7851	-1.55
3.085	-0.2891	-0.7837	-1.56
3.090	-0.2902	-0.7828	-1.57
3.095	-0.2915	-0.7825	-1.57
3.100	-0.2928	-0.7822	-1.56
3.105	-0.2941	-0.7824	-1.56

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.110	-0.2953	-0.7820	-1.56
3.115	-0.2968	-0.7817	-1.55
3.120	-0.2981	-0.7821	-1.55
3.125	-0.2993	-0.7826	-1.54
3.130	-0.3002	-0.7830	-1.53
3.135	-0.3012	-0.7834	-1.52
3.140	-0.3018	-0.7842	-1.51
3.145	-0.3026	-0.7850	-1.50
3.150	-0.3032	-0.7855	-1.49
3.155	-0.3034	-0.7859	-1.49
3.160	-0.3037	-0.7864	-1.48
3.165	-0.3039	-0.7869	-1.48
3.170	-0.3042	-0.7871	-1.47
3.175	-0.3040	-0.7872	-1.47
3.180	-0.3036	-0.7878	-1.47
3.185	-0.3034	-0.7874	-1.48
3.190	-0.3034	-0.7867	-1.48
3.195	-0.3033	-0.7862	-1.49
3.200	-0.3035	-0.7853	-1.50
3.205	-0.3036	-0.7841	-1.51
3.210	-0.3040	-0.7832	-1.51
3.215	-0.3040	-0.7821	-1.52
3.220	-0.3053	-0.7812	-1.53
3.225	-0.3064	-0.7803	-1.53
3.230	-0.3077	-0.7801	-1.53
3.235	-0.3090	-0.7801	-1.52
3.240	-0.3103	-0.7800	-1.52
3.245	-0.3114	-0.7802	-1.51
3.250	-0.3127	-0.7803	-1.51
3.255	-0.3138	-0.7806	-1.50
3.260	-0.3148	-0.7811	-1.49
3.265	-0.3156	-0.7809	-1.49
3.270	-0.3165	-0.7813	-1.48
3.275	-0.3175	-0.7815	-1.48
3.280	-0.3185	-0.7817	-1.47
3.285	-0.3196	-0.7819	-1.47
3.290	-0.3207	-0.7820	-1.46
3.295	-0.3212	-0.7817	-1.46
3.300	-0.3220	-0.7819	-1.46
3.305	-0.3229	-0.7813	-1.46
3.310	-0.3235	-0.7818	-1.45
3.315	-0.3239	-0.7815	-1.45
3.320	-0.3246	-0.7813	-1.45
3.325	-0.3256	-0.7812	-1.45
3.330	-0.3258	-0.7805	-1.45
3.335	-0.3261	-0.7797	-1.46

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.340	-0.3269	-0.7793	-1.46
3.345	-0.3275	-0.7787	-1.47
3.350	-0.3280	-0.7775	-1.47
3.355	-0.3284	-0.7765	-1.48
3.360	-0.3290	-0.7755	-1.49
3.365	-0.3300	-0.7742	-1.50
3.370	-0.3303	-0.7728	-1.51
3.375	-0.3313	-0.7718	-1.52
3.380	-0.3320	-0.7703	-1.53
3.385	-0.3324	-0.7689	-1.54
3.390	-0.3342	-0.7673	-1.55
3.395	-0.3351	-0.7658	-1.56
3.400	-0.3363	-0.7641	-1.57
3.405	-0.3372	-0.7624	-1.58
3.410	-0.3387	-0.7606	-1.59
3.415	-0.3404	-0.7592	-1.60
3.420	-0.3417	-0.7576	-1.61
3.425	-0.3431	-0.7560	-1.62
3.430	-0.3449	-0.7548	-1.62
3.435	-0.3468	-0.7531	-1.63
3.440	-0.3481	-0.7516	-1.64
3.445	-0.3503	-0.7503	-1.64
3.450	-0.3522	-0.7491	-1.64
3.455	-0.3542	-0.7476	-1.65
3.460	-0.3561	-0.7469	-1.65
3.465	-0.3579	-0.7460	-1.65
3.470	-0.3595	-0.7453	-1.64
3.475	-0.3614	-0.7446	-1.64
3.480	-0.3629	-0.7440	-1.64
3.485	-0.3646	-0.7437	-1.64
3.490	-0.3666	-0.7439	-1.62
3.495	-0.3672	-0.7437	-1.62
3.500	-0.3683	-0.7435	-1.62
3.505	-0.3689	-0.7434	-1.62
3.510	-0.3693	-0.7428	-1.62
3.515	-0.3695	-0.7422	-1.63
3.520	-0.3697	-0.7412	-1.64
3.525	-0.3696	-0.7408	-1.64
3.530	-0.3700	-0.7396	-1.65
3.535	-0.3689	-0.7378	-1.67
3.540	-0.3682	-0.7355	-1.70
3.545	-0.3687	-0.7329	-1.72
3.550	-0.3688	-0.7297	-1.75
3.555	-0.3697	-0.7267	-1.77
3.560	-0.3710	-0.7234	-1.80
3.565	-0.3728	-0.7206	-1.82

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.570	-0.3755	-0.7174	-1.83
3.575	-0.3776	-0.7142	-1.85
3.580	-0.3824	-0.7128	-1.84
3.585	-0.3857	-0.7113	-1.84
3.590	-0.3896	-0.7119	-1.81
3.595	-0.3927	-0.7130	-1.79
3.600	-0.3949	-0.7149	-1.76
3.605	-0.3961	-0.7166	-1.74
3.610	-0.3961	-0.7187	-1.72
3.615	-0.3953	-0.7196	-1.71
3.620	-0.3944	-0.7202	-1.71
3.625	-0.3940	-0.7205	-1.71
3.630	-0.3916	-0.7204	-1.72
3.635	-0.3902	-0.7189	-1.74
3.640	-0.3896	-0.7173	-1.76
3.645	-0.3895	-0.7152	-1.78
3.650	-0.3893	-0.7135	-1.80
3.655	-0.3899	-0.7116	-1.81
3.660	-0.3901	-0.7102	-1.83
3.665	-0.3908	-0.7087	-1.84
3.670	-0.3912	-0.7068	-1.85
3.675	-0.3915	-0.7044	-1.88
3.680	-0.3921	-0.7031	-1.88
3.685	-0.3931	-0.7012	-1.90
3.690	-0.3944	-0.6989	-1.91
3.695	-0.3961	-0.6968	-1.92
3.700	-0.3981	-0.6946	-1.93
3.705	-0.4001	-0.6925	-1.94
3.710	-0.4033	-0.6901	-1.95
3.715	-0.4069	-0.6890	-1.94
3.720	-0.4110	-0.6877	-1.93
3.725	-0.4158	-0.6869	-1.91
3.730	-0.4207	-0.6876	-1.87
3.735	-0.4243	-0.6899	-1.83
3.740	-0.4278	-0.6927	-1.79
3.745	-0.4295	-0.6958	-1.75
3.750	-0.4305	-0.6996	-1.71
3.755	-0.4298	-0.7027	-1.69
3.760	-0.4281	-0.7059	-1.66
3.765	-0.4254	-0.7075	-1.67
3.770	-0.4219	-0.7076	-1.68
3.775	-0.4186	-0.7060	-1.72
3.780	-0.4164	-0.7036	-1.75
3.785	-0.4149	-0.6995	-1.80
3.790	-0.4160	-0.6955	-1.83
3.795	-0.4181	-0.6923	-1.84

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.800	-0.4205	-0.6899	-1.85
3.805	-0.4236	-0.6879	-1.85
3.810	-0.4266	-0.6868	-1.85
3.815	-0.4293	-0.6858	-1.84
3.820	-0.4322	-0.6858	-1.82
3.825	-0.4350	-0.6855	-1.81
3.830	-0.4370	-0.6861	-1.79
3.835	-0.4379	-0.6896	-1.76
3.840	-0.4393	-0.6888	-1.76
3.845	-0.4405	-0.6887	-1.75
3.850	-0.4418	-0.6883	-1.75
3.855	-0.4431	-0.6878	-1.74
3.860	-0.4442	-0.6873	-1.74
3.865	-0.4459	-0.6866	-1.74
3.870	-0.4471	-0.6858	-1.74
3.875	-0.4485	-0.6853	-1.73
3.880	-0.4501	-0.6843	-1.73
3.885	-0.4515	-0.6838	-1.73
3.890	-0.4535	-0.6832	-1.72
3.895	-0.4551	-0.6823	-1.72
3.900	-0.4569	-0.6814	-1.72
3.905	-0.4584	-0.6805	-1.72
3.910	-0.4606	-0.6794	-1.71
3.915	-0.4622	-0.6786	-1.71
3.920	-0.4641	-0.6777	-1.71
3.925	-0.4662	-0.6767	-1.71
3.930	-0.4677	-0.6759	-1.70
3.935	-0.4692	-0.6750	-1.70
3.940	-0.4712	-0.6738	-1.70
3.945	-0.4730	-0.6729	-1.70
3.950	-0.4747	-0.6717	-1.70
3.955	-0.4765	-0.6704	-1.70
3.960	-0.4781	-0.6690	-1.70
3.965	-0.4799	-0.6677	-1.70
3.970	-0.4816	-0.6664	-1.70
3.975	-0.4833	-0.6651	-1.70
3.980	-0.4850	-0.6638	-1.70
3.985	-0.4870	-0.6623	-1.70
3.990	-0.4886	-0.6610	-1.70
3.995	-0.4903	-0.6593	-1.71
4.000	-0.4922	-0.6581	-1.71
4.005	-0.4936	-0.6565	-1.71
4.010	-0.4954	-0.6548	-1.71
4.015	-0.4971	-0.6533	-1.71
4.020	-0.4985	-0.6518	-1.72
4.025	-0.5001	-0.6504	-1.72

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.030	-0.5015	-0.6488	-1.72
4.035	-0.5032	-0.6474	-1.72
4.040	-0.5049	-0.6457	-1.73
4.045	-0.5061	-0.6441	-1.73
4.050	-0.5073	-0.6426	-1.74
4.055	-0.5090	-0.6410	-1.74
4.060	-0.5102	-0.6394	-1.74
4.065	-0.5115	-0.6380	-1.75
4.070	-0.5126	-0.6365	-1.75
4.075	-0.5137	-0.6352	-1.76
4.080	-0.5150	-0.6338	-1.76
4.085	-0.5163	-0.6325	-1.76
4.090	-0.5174	-0.6312	-1.76
4.095	-0.5186	-0.6299	-1.77
4.100	-0.5195	-0.6287	-1.77
4.105	-0.5208	-0.6275	-1.77
4.110	-0.5220	-0.6265	-1.77
4.115	-0.5232	-0.6254	-1.77
4.120	-0.5245	-0.6245	-1.77
4.125	-0.5259	-0.6233	-1.77
4.130	-0.5273	-0.6221	-1.77
4.135	-0.5289	-0.6207	-1.77
4.140	-0.5299	-0.6194	-1.77
4.145	-0.5309	-0.6173	-1.79
4.150	-0.5320	-0.6162	-1.79
4.155	-0.5324	-0.6147	-1.80
4.160	-0.5331	-0.6132	-1.80
4.165	-0.5332	-0.6119	-1.81
4.170	-0.5332	-0.6110	-1.82
4.175	-0.5334	-0.6102	-1.83
4.180	-0.5333	-0.6095	-1.83
4.185	-0.5329	-0.6086	-1.84
4.190	-0.5332	-0.6079	-1.85
4.195	-0.5329	-0.6079	-1.85
4.200	-0.5329	-0.6075	-1.85
4.205	-0.5330	-0.6075	-1.85
4.210	-0.5325	-0.6071	-1.86
4.215	-0.5328	-0.6069	-1.86
4.220	-0.5325	-0.6072	-1.86
4.225	-0.5322	-0.6069	-1.86
4.230	-0.5320	-0.6073	-1.86
4.235	-0.5319	-0.6071	-1.86
4.240	-0.5319	-0.6075	-1.86
4.245	-0.5312	-0.6075	-1.86
4.250	-0.5313	-0.6080	-1.86
4.255	-0.5308	-0.6085	-1.86

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.260	-0.5305	-0.6088	-1.86
4.265	-0.5303	-0.6087	-1.86
4.270	-0.5301	-0.6095	-1.85
4.275	-0.5299	-0.6098	-1.85
4.280	-0.5295	-0.6102	-1.85
4.285	-0.5291	-0.6107	-1.85
4.290	-0.5287	-0.6109	-1.85
4.295	-0.5284	-0.6116	-1.85
4.300	-0.5283	-0.6119	-1.85
4.305	-0.5281	-0.6126	-1.84
4.310	-0.5281	-0.6129	-1.84
4.315	-0.5278	-0.6133	-1.84
4.320	-0.5277	-0.6139	-1.84
4.325	-0.5274	-0.6147	-1.83
4.330	-0.5273	-0.6151	-1.83
4.335	-0.5272	-0.6156	-1.82
4.340	-0.5271	-0.6164	-1.82
4.345	-0.5268	-0.6172	-1.81
4.350	-0.5267	-0.6180	-1.81
4.355	-0.5262	-0.6188	-1.81
4.360	-0.5260	-0.6198	-1.80
4.365	-0.5255	-0.6207	-1.79
4.370	-0.5249	-0.6216	-1.79
4.375	-0.5245	-0.6223	-1.79
4.380	-0.5239	-0.6233	-1.78
4.385	-0.5235	-0.6242	-1.78
4.390	-0.5226	-0.6250	-1.78
4.395	-0.5223	-0.6256	-1.78
4.400	-0.5217	-0.6263	-1.77
4.405	-0.5210	-0.6271	-1.77
4.410	-0.5205	-0.6275	-1.77
4.415	-0.5197	-0.6277	-1.78
4.420	-0.5195	-0.6281	-1.78
4.425	-0.5188	-0.6286	-1.78
4.430	-0.5186	-0.6289	-1.78
4.435	-0.5183	-0.6289	-1.78
4.440	-0.5181	-0.6292	-1.78
4.445	-0.5180	-0.6296	-1.77
4.450	-0.5178	-0.6298	-1.77
4.455	-0.5180	-0.6305	-1.77
4.460	-0.5178	-0.6309	-1.76
4.465	-0.5173	-0.6317	-1.76
4.470	-0.5169	-0.6328	-1.75
4.475	-0.5167	-0.6336	-1.75
4.480	-0.5161	-0.6353	-1.74
4.485	-0.5163	-0.6366	-1.73

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.490	-0.5159	-0.6383	-1.72
4.495	-0.5159	-0.6394	-1.71
4.500	-0.5153	-0.6408	-1.70
4.505	-0.5147	-0.6415	-1.70
4.510	-0.5143	-0.6420	-1.70
4.515	-0.5137	-0.6424	-1.70
4.520	-0.5128	-0.6428	-1.70
4.525	-0.5125	-0.6430	-1.70
4.530	-0.5120	-0.6424	-1.71
4.535	-0.5119	-0.6416	-1.71
4.540	-0.5121	-0.6412	-1.72
4.545	-0.5123	-0.6404	-1.72
4.550	-0.5125	-0.6401	-1.72
4.555	-0.5124	-0.6395	-1.73
4.560	-0.5130	-0.6391	-1.73
4.565	-0.5130	-0.6386	-1.73
4.570	-0.5135	-0.6379	-1.74
4.575	-0.5139	-0.6373	-1.74
4.580	-0.5139	-0.6362	-1.75
4.585	-0.5146	-0.6354	-1.75
4.590	-0.5147	-0.6346	-1.75
4.595	-0.5150	-0.6336	-1.76
4.600	-0.5154	-0.6326	-1.77
4.605	-0.5158	-0.6319	-1.77
4.610	-0.5164	-0.6311	-1.77
4.615	-0.5168	-0.6303	-1.78
4.620	-0.5169	-0.6290	-1.79
4.625	-0.5172	-0.6282	-1.79
4.630	-0.5178	-0.6276	-1.79
4.635	-0.5181	-0.6267	-1.80
4.640	-0.5183	-0.6259	-1.80
4.645	-0.5186	-0.6248	-1.81
4.650	-0.5187	-0.6241	-1.81
4.655	-0.5189	-0.6234	-1.82
4.660	-0.5190	-0.6228	-1.82
4.665	-0.5193	-0.6223	-1.82
4.670	-0.5189	-0.6214	-1.83
4.675	-0.5187	-0.6206	-1.84
4.680	-0.5185	-0.6201	-1.85
4.685	-0.5182	-0.6194	-1.86
4.690	-0.5179	-0.6187	-1.86
4.695	-0.5179	-0.6184	-1.87
4.700	-0.5176	-0.6178	-1.87
4.705	-0.5171	-0.6169	-1.88
4.710	-0.5177	-0.6161	-1.89
4.715	-0.5176	-0.6152	-1.89

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.720	-0.5180	-0.6141	-1.90
4.725	-0.5185	-0.6130	-1.91
4.730	-0.5187	-0.6118	-1.92
4.735	-0.5194	-0.6107	-1.92
4.740	-0.5201	-0.6092	-1.93
4.745	-0.5207	-0.6074	-1.94
4.750	-0.5219	-0.6065	-1.94
4.755	-0.5227	-0.6048	-1.95
4.760	-0.5235	-0.6035	-1.95
4.765	-0.5236	-0.6026	-1.96
4.770	-0.5242	-0.6017	-1.96
4.775	-0.5243	-0.6006	-1.97
4.780	-0.5245	-0.6002	-1.97
4.785	-0.5239	-0.5994	-1.98
4.790	-0.5235	-0.5983	-1.99
4.795	-0.5227	-0.5977	-2.00
4.800	-0.5220	-0.5974	-2.01
4.805	-0.5212	-0.5965	-2.02
4.810	-0.5202	-0.5954	-2.04
4.815	-0.5193	-0.5951	-2.05
4.820	-0.5185	-0.5951	-2.06
4.825	-0.5174	-0.5947	-2.07
4.830	-0.5171	-0.5943	-2.07
4.835	-0.5168	-0.5942	-2.08
4.840	-0.5163	-0.5943	-2.08
4.845	-0.5165	-0.5939	-2.08
4.850	-0.5168	-0.5945	-2.07
4.855	-0.5166	-0.5950	-2.07
4.860	-0.5167	-0.5949	-2.07
4.865	-0.5173	-0.5958	-2.06
4.870	-0.5177	-0.5960	-2.05
4.875	-0.5181	-0.5964	-2.05
4.880	-0.5184	-0.5969	-2.04
4.885	-0.5187	-0.5976	-2.03
4.890	-0.5196	-0.5979	-2.02
4.895	-0.5201	-0.5986	-2.01
4.900	-0.5203	-0.5988	-2.01
4.905	-0.5210	-0.5991	-2.00
4.910	-0.5215	-0.5992	-2.00
4.915	-0.5221	-0.5996	-1.99
4.920	-0.5227	-0.5998	-1.99
4.925	-0.5232	-0.6002	-1.98
4.930	-0.5238	-0.6006	-1.97
4.935	-0.5243	-0.6007	-1.97
4.940	-0.5249	-0.6009	-1.96
4.945	-0.5254	-0.6011	-1.96

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.950	-0.5261	-0.6013	-1.95
4.955	-0.5266	-0.6013	-1.95
4.960	-0.5270	-0.6014	-1.94
4.965	-0.5275	-0.6015	-1.94
4.970	-0.5280	-0.6016	-1.93
4.975	-0.5283	-0.6018	-1.93
4.980	-0.5290	-0.6019	-1.92
4.985	-0.5292	-0.6022	-1.92
4.990	-0.5297	-0.6023	-1.92
4.995	-0.5299	-0.6025	-1.91
5.000	-0.5305	-0.6028	-1.91
5.005	-0.5308	-0.6030	-1.90
5.010	-0.5313	-0.6030	-1.90
5.015	-0.5316	-0.6034	-1.89
5.020	-0.5324	-0.6033	-1.89
5.025	-0.5323	-0.6035	-1.89
5.030	-0.5330	-0.6029	-1.89
5.035	-0.5329	-0.6030	-1.89
5.040	-0.5330	-0.6021	-1.89
5.045	-0.5326	-0.6027	-1.89
5.050	-0.5325	-0.6019	-1.90
5.055	-0.5323	-0.6025	-1.90
5.060	-0.5314	-0.6016	-1.91
5.065	-0.5316	-0.6023	-1.90
5.070	-0.5312	-0.6022	-1.91
5.075	-0.5322	-0.6026	-1.90
5.080	-0.5317	-0.6027	-1.90
5.085	-0.5329	-0.6026	-1.89
5.090	-0.5329	-0.6027	-1.89
5.095	-0.5342	-0.6022	-1.88
5.100	-0.5347	-0.6023	-1.88
5.105	-0.5355	-0.6015	-1.88
5.110	-0.5365	-0.6017	-1.87
5.115	-0.5376	-0.6012	-1.87
5.120	-0.5385	-0.6008	-1.86
5.125	-0.5399	-0.6001	-1.86
5.130	-0.5409	-0.5996	-1.86
5.135	-0.5420	-0.5990	-1.85
5.140	-0.5428	-0.5980	-1.86
5.145	-0.5441	-0.5967	-1.86
5.150	-0.5452	-0.5960	-1.85
5.155	-0.5466	-0.5948	-1.85
5.160	-0.5478	-0.5936	-1.85
5.165	-0.5489	-0.5921	-1.86
5.170	-0.5503	-0.5904	-1.86
5.175	-0.5514	-0.5887	-1.87

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.180	-0.5527	-0.5871	-1.87
5.185	-0.5541	-0.5852	-1.87
5.190	-0.5548	-0.5830	-1.89
5.195	-0.5561	-0.5813	-1.89
5.200	-0.5572	-0.5793	-1.90
5.205	-0.5583	-0.5771	-1.91
5.210	-0.5592	-0.5754	-1.91
5.215	-0.5600	-0.5732	-1.92
5.220	-0.5612	-0.5709	-1.93
5.225	-0.5621	-0.5686	-1.94
5.230	-0.5631	-0.5665	-1.95
5.235	-0.5640	-0.5643	-1.96
5.240	-0.5650	-0.5622	-1.97
5.245	-0.5657	-0.5600	-1.98
5.250	-0.5667	-0.5579	-1.99
5.255	-0.5677	-0.5558	-2.00
5.260	-0.5685	-0.5535	-2.01
5.265	-0.5698	-0.5516	-2.01
5.270	-0.5710	-0.5497	-2.02
5.275	-0.5715	-0.5476	-2.03
5.280	-0.5725	-0.5457	-2.04
5.285	-0.5734	-0.5440	-2.04
5.290	-0.5742	-0.5420	-2.05
5.295	-0.5752	-0.5402	-2.06
5.300	-0.5760	-0.5386	-2.06
5.305	-0.5769	-0.5374	-2.07
5.310	-0.5777	-0.5355	-2.07
5.315	-0.5785	-0.5336	-2.08
5.320	-0.5791	-0.5320	-2.09
5.325	-0.5798	-0.5304	-2.09
5.330	-0.5804	-0.5289	-2.10
5.335	-0.5810	-0.5272	-2.11
5.340	-0.5814	-0.5257	-2.12
5.345	-0.5819	-0.5241	-2.12
5.350	-0.5827	-0.5224	-2.13
5.355	-0.5833	-0.5207	-2.14
5.360	-0.5838	-0.5191	-2.14
5.365	-0.5842	-0.5171	-2.16
5.370	-0.5850	-0.5154	-2.16
5.375	-0.5859	-0.5139	-2.17
5.380	-0.5866	-0.5123	-2.17
5.385	-0.5875	-0.5104	-2.18
5.390	-0.5884	-0.5089	-2.18
5.395	-0.5894	-0.5077	-2.18
5.400	-0.5903	-0.5064	-2.18
5.405	-0.5911	-0.5050	-2.19

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.410	-0.5921	-0.5038	-2.19
5.415	-0.5932	-0.5024	-2.19
5.420	-0.5939	-0.5015	-2.19
5.425	-0.5946	-0.4999	-2.19
5.430	-0.5955	-0.4991	-2.19
5.435	-0.5962	-0.4979	-2.19
5.440	-0.5970	-0.4969	-2.19
5.445	-0.5977	-0.4960	-2.19
5.450	-0.5987	-0.4954	-2.19
5.455	-0.5990	-0.4948	-2.19
5.460	-0.6001	-0.4937	-2.19
5.465	-0.6009	-0.4926	-2.19
5.470	-0.6010	-0.4920	-2.20
5.475	-0.6019	-0.4911	-2.19
5.480	-0.6022	-0.4904	-2.20
5.485	-0.6030	-0.4896	-2.20
5.490	-0.6037	-0.4888	-2.19
5.495	-0.6048	-0.4880	-2.19
5.500	-0.6057	-0.4872	-2.19
5.505	-0.6068	-0.4869	-2.18
5.510	-0.6074	-0.4865	-2.18
5.515	-0.6085	-0.4864	-2.17
5.520	-0.6093	-0.4863	-2.16
5.525	-0.6100	-0.4862	-2.16
5.530	-0.6113	-0.4862	-2.15
5.535	-0.6121	-0.4862	-2.14
5.540	-0.6125	-0.4863	-2.13
5.545	-0.6133	-0.4864	-2.13
5.550	-0.6139	-0.4865	-2.12
5.555	-0.6145	-0.4866	-2.12
5.560	-0.6153	-0.4865	-2.11
5.565	-0.6160	-0.4867	-2.10
5.570	-0.6168	-0.4866	-2.10
5.575	-0.6176	-0.4866	-2.09
5.580	-0.6183	-0.4867	-2.08
5.585	-0.6192	-0.4866	-2.07
5.590	-0.6201	-0.4867	-2.07
5.595	-0.6211	-0.4867	-2.06
5.600	-0.6222	-0.4866	-2.05
5.605	-0.6232	-0.4868	-2.04
5.610	-0.6240	-0.4871	-2.03
5.615	-0.6249	-0.4874	-2.02
5.620	-0.6260	-0.4876	-2.01
5.625	-0.6270	-0.4879	-2.00
5.630	-0.6276	-0.4883	-1.99
5.635	-0.6284	-0.4885	-1.98

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.640	-0.6291	-0.4884	-1.98
5.645	-0.6297	-0.4890	-1.97
5.650	-0.6303	-0.4890	-1.96
5.655	-0.6308	-0.4893	-1.96
5.660	-0.6315	-0.4894	-1.95
5.665	-0.6321	-0.4893	-1.94
5.670	-0.6325	-0.4894	-1.94
5.675	-0.6332	-0.4893	-1.94
5.680	-0.6336	-0.4897	-1.93
5.685	-0.6340	-0.4896	-1.93
5.690	-0.6341	-0.4892	-1.93
5.695	-0.6350	-0.4893	-1.92
5.700	-0.6352	-0.4888	-1.92
5.705	-0.6352	-0.4884	-1.92
5.710	-0.6358	-0.4888	-1.92
5.715	-0.6358	-0.4883	-1.92
5.720	-0.6363	-0.4882	-1.92
5.725	-0.6366	-0.4880	-1.92
5.730	-0.6367	-0.4879	-1.91
5.735	-0.6371	-0.4874	-1.92
5.740	-0.6370	-0.4869	-1.92
5.745	-0.6372	-0.4863	-1.92
5.750	-0.6373	-0.4859	-1.92
5.755	-0.6374	-0.4855	-1.92
5.760	-0.6375	-0.4848	-1.93
5.765	-0.6376	-0.4846	-1.93
5.770	-0.6376	-0.4833	-1.94
5.775	-0.6375	-0.4827	-1.94
5.780	-0.6378	-0.4818	-1.95
5.785	-0.6376	-0.4809	-1.95
5.790	-0.6378	-0.4797	-1.96
5.795	-0.6376	-0.4787	-1.97
5.800	-0.6373	-0.4779	-1.98
5.805	-0.6372	-0.4766	-1.98
5.810	-0.6368	-0.4755	-2.00
5.815	-0.6370	-0.4744	-2.00
5.820	-0.6370	-0.4733	-2.01
5.825	-0.6368	-0.4718	-2.02
5.830	-0.6366	-0.4708	-2.03
5.835	-0.6365	-0.4694	-2.04
5.840	-0.6364	-0.4681	-2.05
5.845	-0.6363	-0.4669	-2.06
5.850	-0.6361	-0.4654	-2.07
5.855	-0.6363	-0.4641	-2.07
5.860	-0.6362	-0.4626	-2.08
5.865	-0.6365	-0.4613	-2.09

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.870	-0.6365	-0.4598	-2.10
5.875	-0.6367	-0.4587	-2.11
5.880	-0.6369	-0.4575	-2.11
5.885	-0.6370	-0.4561	-2.12
5.890	-0.6372	-0.4551	-2.12
5.895	-0.6376	-0.4544	-2.13
5.900	-0.6375	-0.4532	-2.13
5.905	-0.6379	-0.4520	-2.14
5.910	-0.6378	-0.4515	-2.14
5.915	-0.6378	-0.4506	-2.15
5.920	-0.6376	-0.4496	-2.16
5.925	-0.6376	-0.4487	-2.16
5.930	-0.6373	-0.4476	-2.17
5.935	-0.6371	-0.4468	-2.18
5.940	-0.6374	-0.4457	-2.18
5.945	-0.6373	-0.4447	-2.19
5.950	-0.6372	-0.4435	-2.20
5.955	-0.6376	-0.4424	-2.20
5.960	-0.6377	-0.4415	-2.21
5.965	-0.6379	-0.4403	-2.21
5.970	-0.6384	-0.4399	-2.21
5.975	-0.6387	-0.4392	-2.21
5.980	-0.6397	-0.4390	-2.20
5.985	-0.6399	-0.4385	-2.21
5.990	-0.6404	-0.4388	-2.20
5.995	-0.6409	-0.4385	-2.20
6.000	-0.6414	-0.4390	-2.19

2 Measurement Quality

Reliable band: from 45 MHz (S11 error σ = 1.0e-03); the points below it are kept in the plot (shaded, outside the scale) and in the tables, but are not reliable.
Spike removal: on (isolated point, 2nd difference > 6 x local noise and > S11 error budget, not shared with another sweep); replaced 1 point(s): Short at 4885 MHz ($|\Delta S11|$ = 1.4e-03).
Outliers shared by several sweeps, left alone (instrument artefacts cancelled by the calibration): 3835 MHz.

- Unreliable below 45 MHz: 8 point(s) (3 mm probe, S11 error σ = 1.0e-03). There the reference liquids resolve ε_r only to ±17.2 at 5 MHz … ±2.1 at 40 MHz per σ (usable: $\leq$ ±2). Values kept, but left out of the chart scale and of the summary medians; start the sweep at $\geq$ 45 MHz.

- Gn shows no radiation signature even at the top of the sweep (3 GHz – 6 GHz): mean arg(Gn) = -108° (radiation: -90°, median deviation 20°) and $|Gn| \propto f^{0.4}$ (radiation: f^3). Over the whole sweep it holds noise and the inconsistency between the reference liquids instead (its term is ≈7% of the Ref 1 (Distilled Water) admittance), so the full method's radiation correction is empirical here: compare with the simplified cross-check, especially for samples whose ε_r lies outside the range of the reference liquids.

- 2 frequency point(s) excluded: reference liquids nearly indistinguishable ($|S11_ref1 - S11_ref2| < 1e\text{-}02$). Consider starting the sweep above the affected range.